

Recapitulation, 11/10 2025:

$$\mathcal{F}(m) = \mathcal{F}_0 + a_2 m^2 + a_4 m^4 + \text{higher order terms} \Rightarrow \begin{array}{l} m = 0 \\ m = -\sqrt{t\bar{a}_2/2a_4} \\ m = +\sqrt{t\bar{a}_2/2a_4} \end{array} \quad \begin{array}{l} \text{Hence} \\ \beta = 1/2 \end{array}$$

$$\mathcal{H}_N = -J \sum_{i=0}^{N-1} s_i s_{i+1} - h \sum_{i=0}^{N-1} s_i \Rightarrow \text{Domain size} \sim \exp(+2\beta J) \rightarrow \infty \text{ with } T \rightarrow 0$$

→ no phase transition

$$H = - \sum_{nn} \delta(s_i - s_j) \Rightarrow \beta \left(\frac{F(s)}{N} - \frac{F(s=0)}{N} \right) = \frac{q-1}{2q} \left(q - \frac{\beta}{2} \right) s^2 - \frac{1}{6} (q-1)(q-2) s^3 + \dots$$

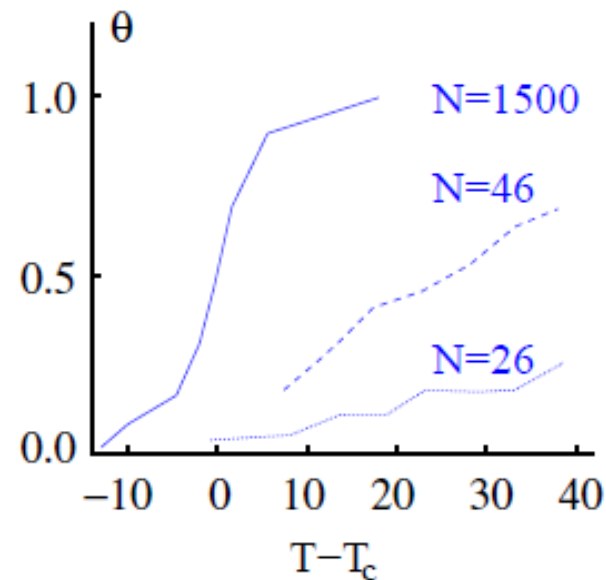
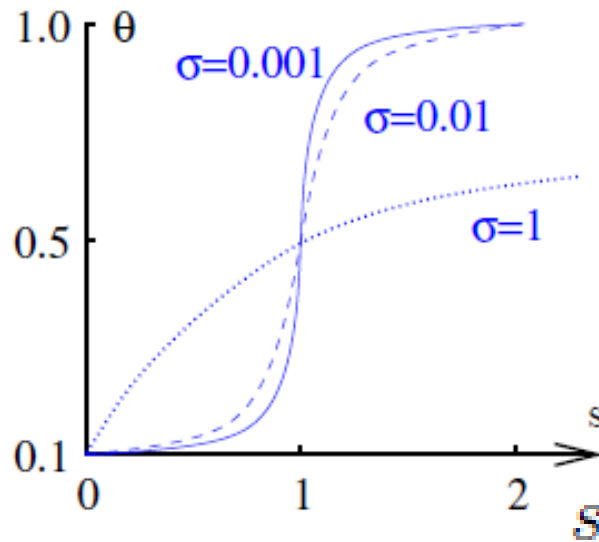


1'st order

Low energy > high entropy, double well

Helix-coil transition with system size

$$\text{domain size} = 1/\sigma \sim \exp(-\beta J)$$



$$\Theta = \frac{s}{2\lambda_{max}} \left(1 + \frac{s + 2\sigma - 1}{\sqrt{(1-s)^2 + 4\sigma s}} \right)$$

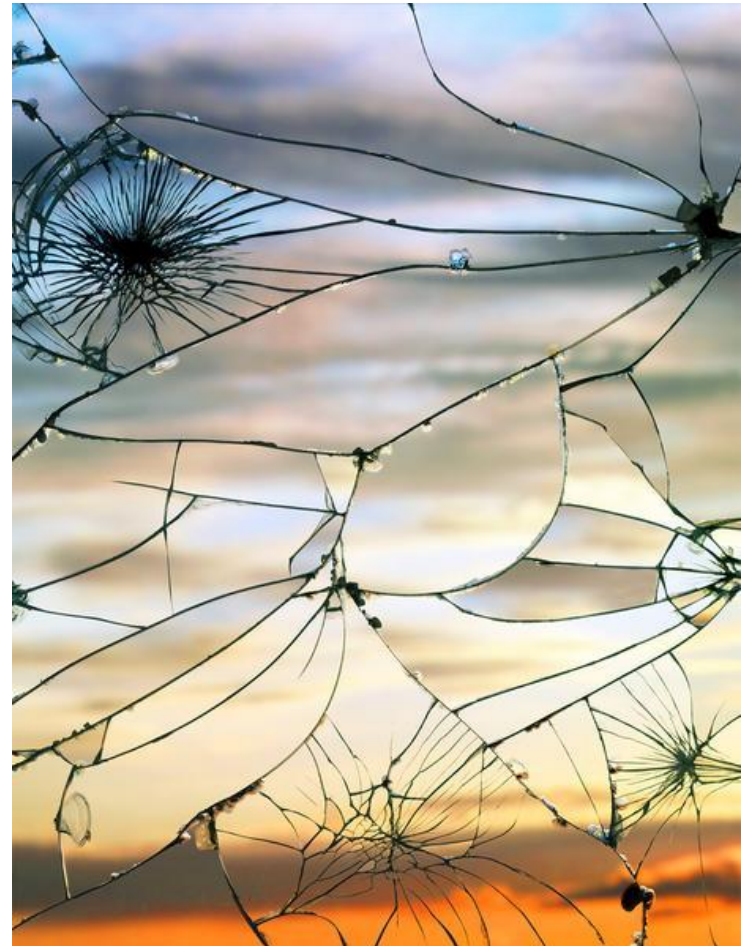


Transition, but not a phase transition.... Small $\sigma \rightarrow 0$ with $T \rightarrow 0$ (cost of boundary high at $T=0$)

Right: The used data is the temperature-induced coil-to-helix transition of the polypeptide poly- γ -benzyl-L-glutamate in mixed dichloroacetic acid-ethylene dichloride solvent (Zimm, Doty and Iso (1959)). Usually the helix fraction would decrease with T , but for this polymer the helix stabilizing forces increases more with temperature than the contribution from the entropy.

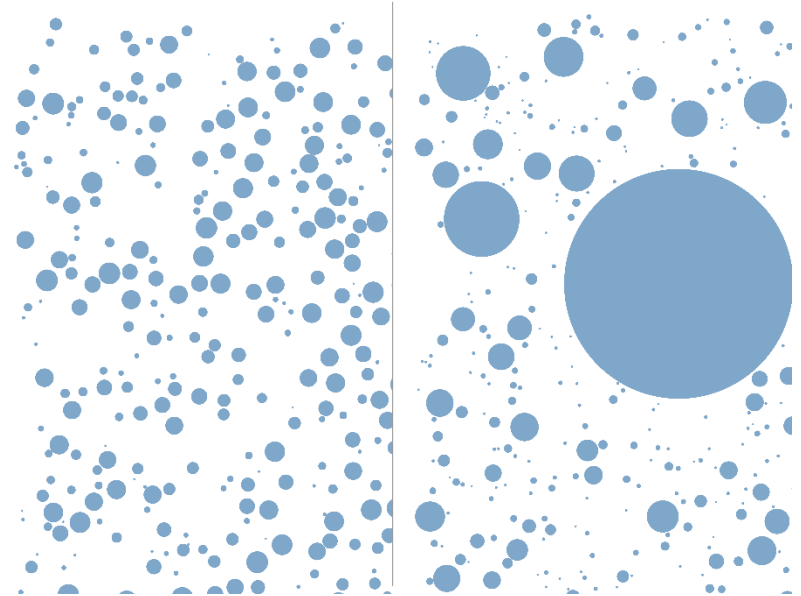
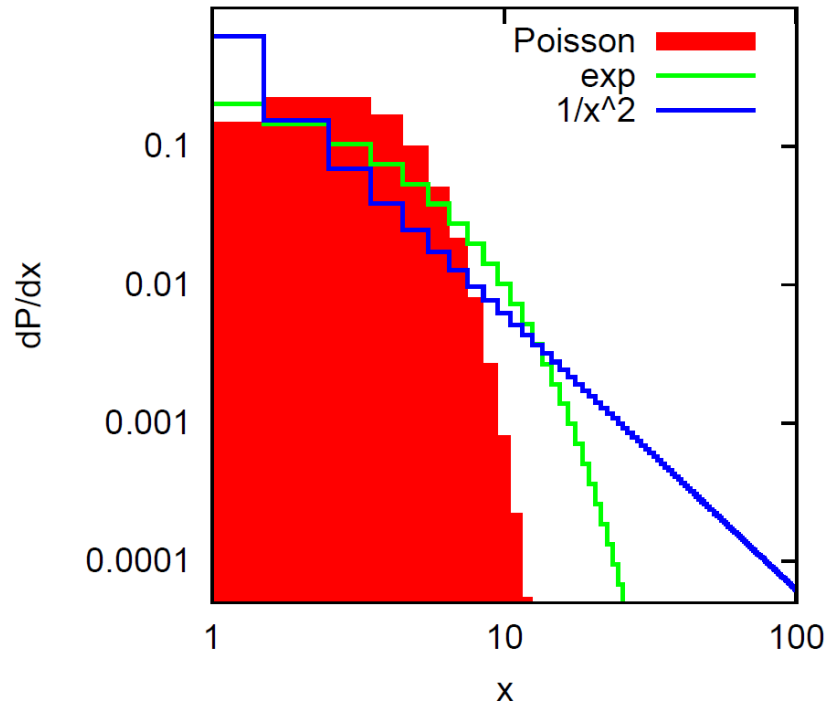
Percolation Fractals & Fragmentation

Kim Sneppen, September 2025



Power laws are different:

Notice log-log!!!



$$\begin{aligned}
 P_{scale}(k) &= \exp(-k/s) & \text{with} & & P_{power-law}(k) &= k^{-\gamma} \\
 P_{scale}(s) &= \exp(-1) & \text{with} & & P_{power-law}(s) &= s^{-\gamma} \\
 \frac{P_{scale}(100 \cdot s)}{P_{scale}(50 \cdot s)} &= \exp(-50) & \text{with} & & \frac{P_{power-law}(100 \cdot s)}{P_{power-law}(50 \cdot s)} &= 2^{-\gamma}
 \end{aligned}$$

You pattern disappear beyond the scale s

You repeat pattern on any scale

Scale free Power laws

$$\begin{aligned}P(s \cdot k)/P(k) &= P(s)/P(1) \\ \Rightarrow \log(P(s \cdot k)/P(1)) &= \log(P(k)/P(1)) + \log(P(s)/P(1)) \\ \Rightarrow f(s \cdot k) &= f(s) + f(k)\end{aligned}$$

where $f(k) = \log(P(k)/P(1))$ and $f(1) = 0$. Thus $f(k)$ is a logarithm of k :

$$\log(P(k)/P(1)) = \gamma \cdot \log(k) \Rightarrow P(k) \propto k^\gamma \quad (2.3)$$

Reversely, if one assumes that $n(k) \propto k^\gamma$, “scale-freeness” is proven by multiplying the argument k with a factor s , and observing that the frequency n changes with the same factor s^γ for all values (scales) of k .

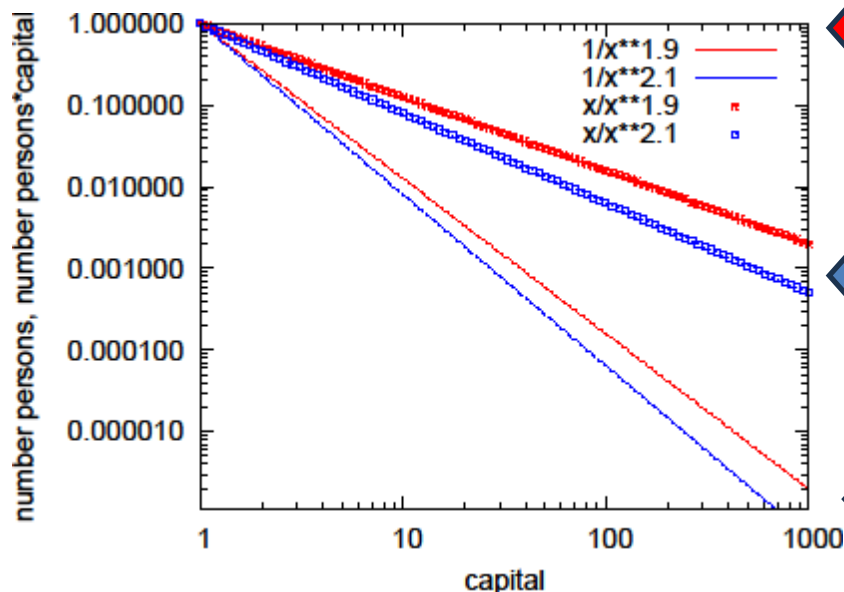
Power law are everywhere:

- Earthquakes can be very very large, even if most of the recorded earthquakes are so small that we wouldn't even feel them without sensitive equipment.
- Solar flares can be huge, although most solar activity is fairly limited. (The “Carrington event” in 1859 gave rise to aurora all over USA, at a level where the illuminated night sky allowed people to read.) The probability to obtain an Energy E is estimated to scale as $P(E) \sim 1/E^{1.6 \pm 0.06}$ for solar x-ray bursts [11].
- Most people have a few 100 followers on their internet activity, but some have millions.
- Gene regulatory networks, where most proteins only associate to few others, but some are central hubs and associate with most others.
- Turbulent liquids can have large vortexes with huge velocity gradients, even if most of the liquid is locally laminar.
- Financial crashes can become very big, even if typical day-to-day stock market fluctuations are small.

Power law scaling $1/x^2$ is marginal:

- Mini Question:

How is richness distribution $1/\$^{1.9}$
different from $1/\$^{2.1}$



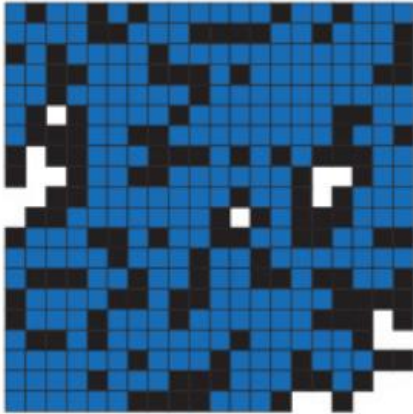
Probability that
A \$ is found in person
With capital $>x$

Probability that
A 1\$ is found in person
with capital x

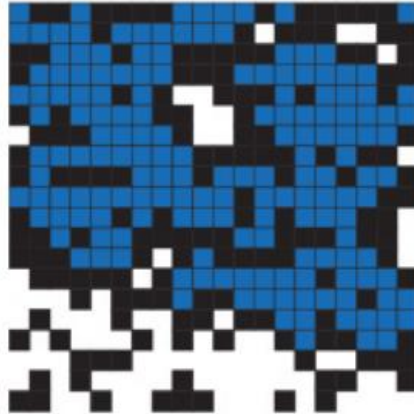
Probability that
a given person
has capital x

Percolation

$p = 0.65$



$p = 0.60$



$p = 0.55$



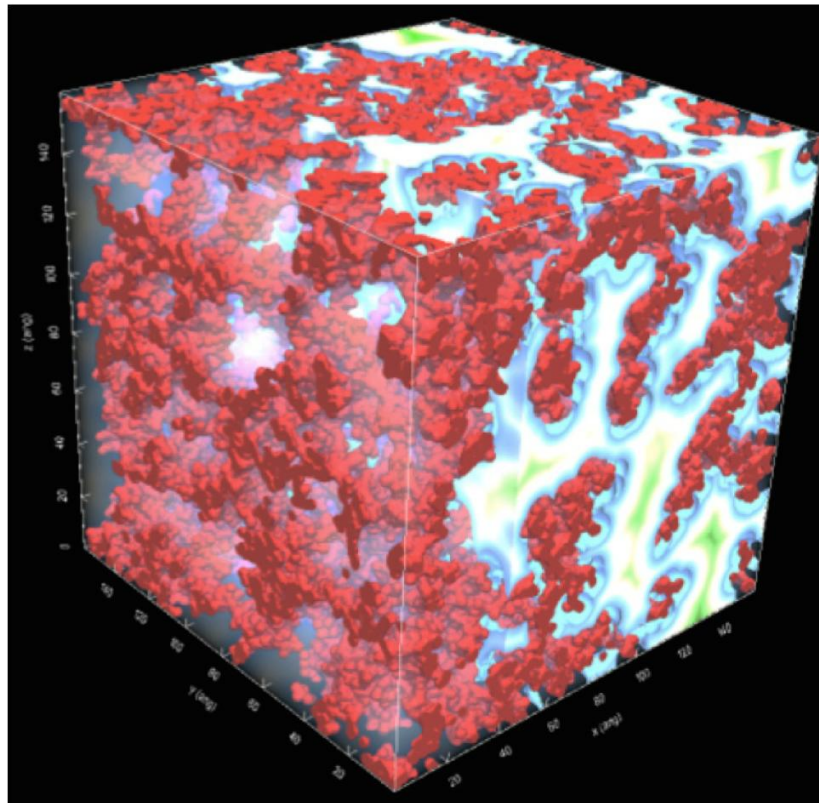
Percolation is less probable as the site vacancy probability p decreases

Look at **blue sites** (water carrying), can one transport water
Across the structure

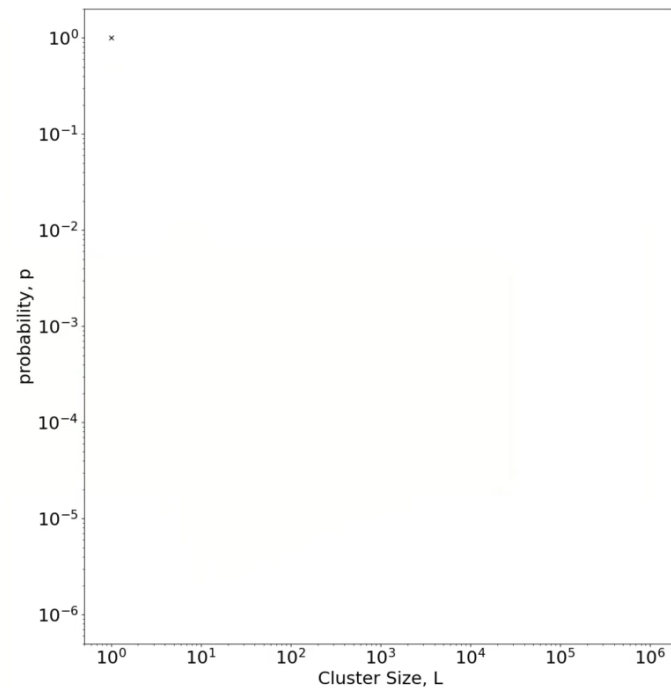
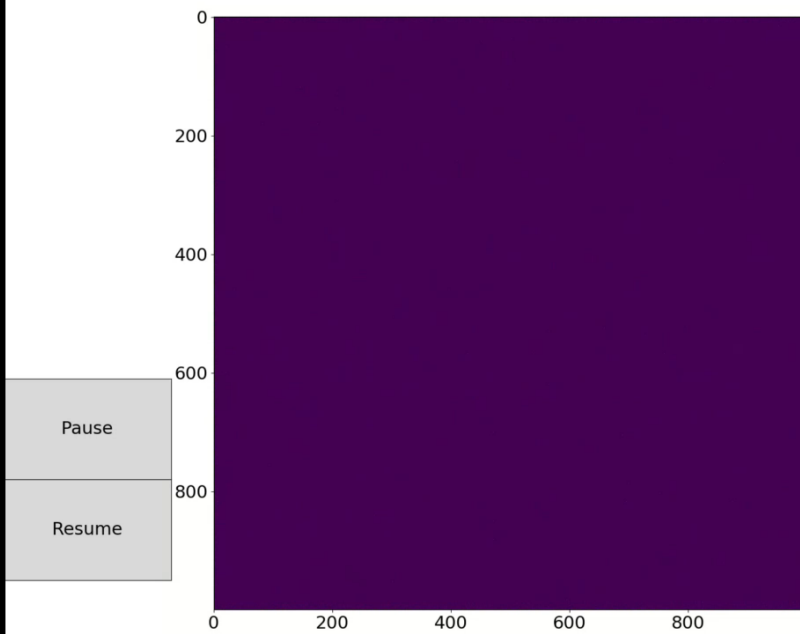
Important for, for example, oil recovery,... pores should percolate
To allow oil to be drained from sediments.

Important for pollution, if underground percolate then dangerous
chemicals may spread.

Percolation, real picture:



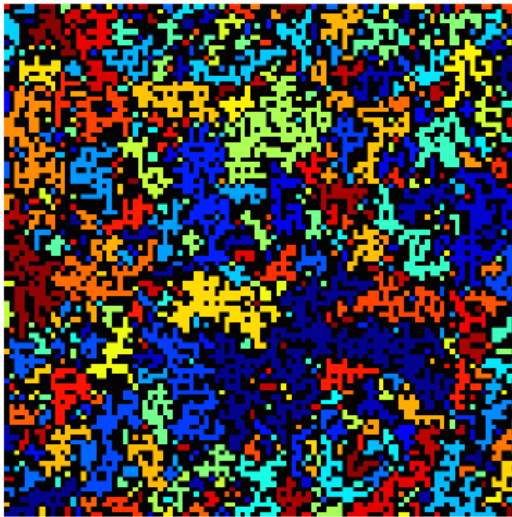
Marcus Engsig simulation (*increasing p*):



Probability 0

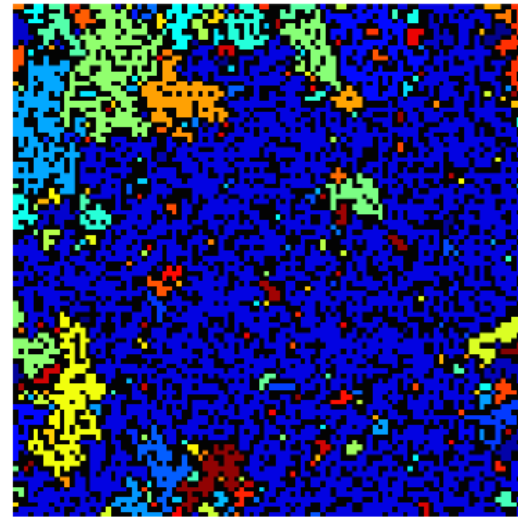
Object of study: Connected Clusters

$p=0.5$



Small p ,
many small clusters

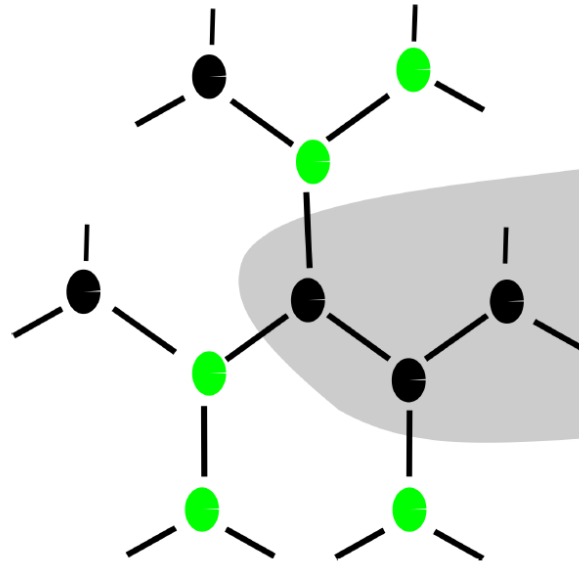
$p=0.6$



Slightly larger p ,
One big cluster

→ **A threshold: The Percolation threshold where there are**
→ **clusters of all sizes**

Simplified lattice---without any loops: Bethe Lattice:



Advantage, can be solved analytically because one broken bond allow us to disregard all that's behind this bond

Mini tutorial:

How many directed paths are there between two points in a Bethe-lattice?

Bethe lattice, dimensionality

$$N(\leq x) = 1 + 3 \cdot (1 + 2 + 2^2 + \dots + 2^{x-1}) = 1 + 3 \cdot \sum_{i=0}^{x-1} 2^i = 3 \cdot 2^x - 2$$

This exponential is faster growing with x than any finite dimension D ,

Remember if dimension is D then: $N_D(\leq x) \propto x^D$

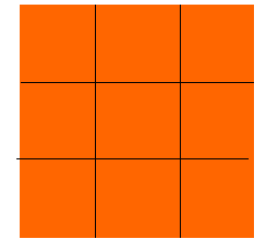
Thus N for Bethe lattice grows faster than any finite dimension D :

$$2^x \gg x^D$$

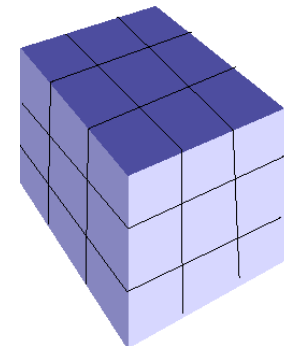
→ dimension D of the Bethe lattice is infinite!



$N=L$



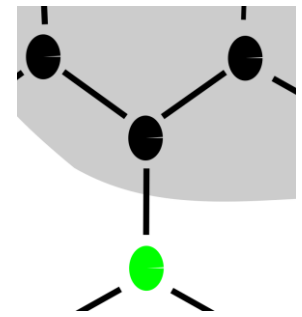
$N=L^2$



$N=L^3$

Critical point:

- Critical branching occur when each new active site connects to in average one new active site
- If probability to be active is p , and each point have $z=3$ neighbors:



$$p_c \cdot (z - 1) = 1, \quad \text{hence} \quad p_c = \frac{1}{z - 1} = \frac{1}{2}$$

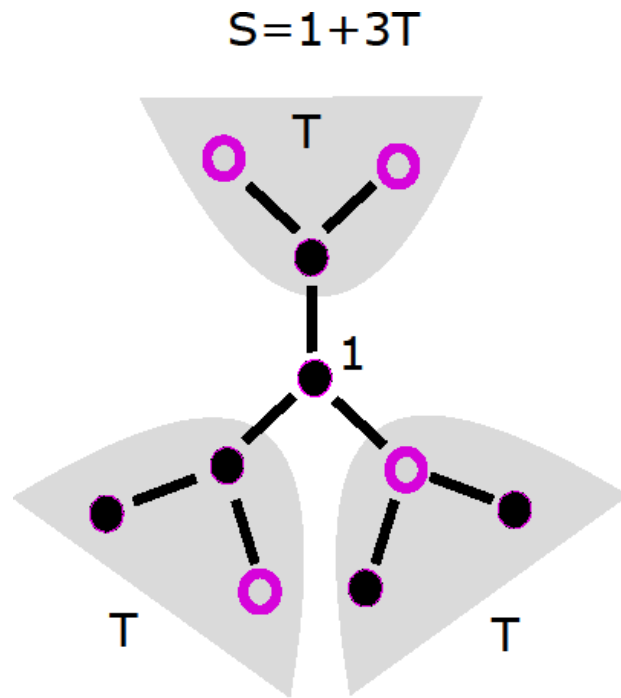
Tutorial:

Mini tutorial:

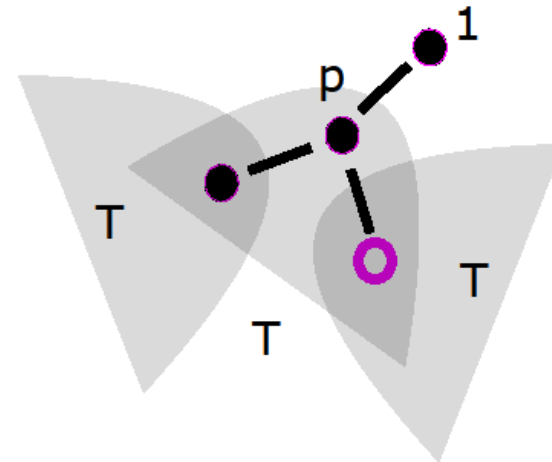
When a quantity $\mathcal{Q}$ scales as $\mathcal{Q} \sim (p_c - p)^{-\gamma}$, does a larger value of γ then imply that $\mathcal{Q}$ tends to become relatively larger or smaller when one approaches the critical point p_c , that is, when the limit $\lim_{p \rightarrow p_c} \mathcal{Q}$ is approached?

Clusters $\rightarrow$ Mean cluster size

- Depend on occupation probability p
- Recursion relation:



$$T = p(1 + 2T)$$



First critical exponent:

---average size of cluster *starting on a point* on the cluster

From last page

$$T = \frac{p}{1 - 2p} \quad \text{for } p < p_c = 1/2$$

$$S(p) = 1 + 3T = \frac{1 + p}{2(p_c - p)} \propto (p_c - p)^{-1}$$

$$S(p) \propto |p_c - p|^{-\gamma} \quad \text{With } \gamma = 1$$

I.e. when one **half distance** to p_c one **double average** cluster
.... *How much more order with change of p (χ in Ising model...
= dm/dh)*

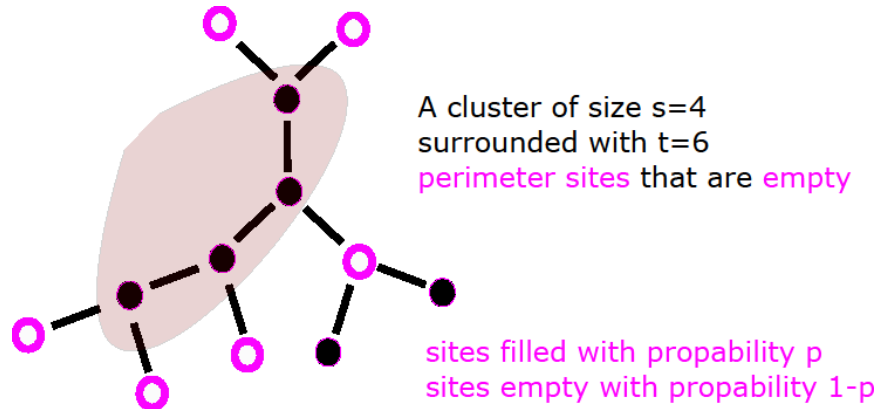
Recapitulation 16/9 2025

- Power law $\leftrightarrow$ Scale free
- Percolation... $p > p_c$ connected path across a system
- Percolation $p = p_c$ fragments of all sizes $\rightarrow$ scale free
- Bethe lattice, no loops, infinite dimensional
- Average cluster size from occupied point: $S(p) = 1/(p_c - p)$

Number of clusters of size s and periphery length t

$$n_s(p) = \sum_t g_{s,t} p^s (1-p)^t,$$

Where $g(s,t)$ is number of different cluster types
of size s and perimeter t



We will not calculate $g(s,t)$, but just proof that $t=s+2$.
i.e. $g(s,t)=0$ if t is not equal $s+2$.

Iterative Proof for $t=s+2$

- For $s=1$, then $t=3$
- For **each added site** (on boundary) one loose 1 perimeter site (to cluster) and win 2 new perimeter sites:
- **$s=s+1$** and **$t=t-1+2=t+1$** $\rightarrow$ again one have $t=s+2$

...we don't need to know $g(s, s+2)$
to determine the scaling:

$$n_s(p) = \sum g_{s,t} p^s (1-p)^t$$

$$n_s(p) = g_{s,2+s} \cdot p^s \cdot (1-p)^{2+s}$$

Only consider Ratio

$$\begin{aligned} \frac{n_s(p)}{n_s(p_c)} &= \left(\frac{1-p}{1-p_c} \right)^2 \left(\frac{p}{p_c} \cdot \frac{1-p}{1-p_c} \right)^s \\ &= \left(\frac{1-p}{1-p_c} \right)^2 \exp \left(s \cdot \ln \left(\frac{p}{p_c} \cdot \frac{1-p}{1-p_c} \right) \right) \end{aligned}$$

$$\stackrel{p_c=1/2}{=} \left(\frac{1-p}{1-p_c} \right)^2 \exp \left(s \cdot \ln(4p - 4p^2) \right) .$$

$$\sim \left(\frac{1-p}{1-p_c} \right)^2 \cdot \exp \left(s \cdot \ln(1 - 4 \cdot (p - p_c)^2) \right)$$

$$\sim \left(\frac{1-p}{1-p_c} \right)^2 \cdot \exp \left(-\frac{s}{s_{p-p_c}} \right)$$

$$s_{p-p_c} \equiv \frac{-1}{\ln(1 - 4 \cdot (p - p_c)^2)} \sim \frac{1}{4} \cdot (p - p_c)^{-2} \propto |p - p_c|^{-1/\sigma}$$

Taylor expand around:

$$p_c = 1/2$$

$$4p - 4p^2 = 1 - 4 \cdot (p - p_c)^2$$

$$p_c = 1/2$$

Taylor expand around:

$$p_c = 1/2$$

$$4p - 4p^2 = 1 - 4 \cdot (p - p_c)^2$$

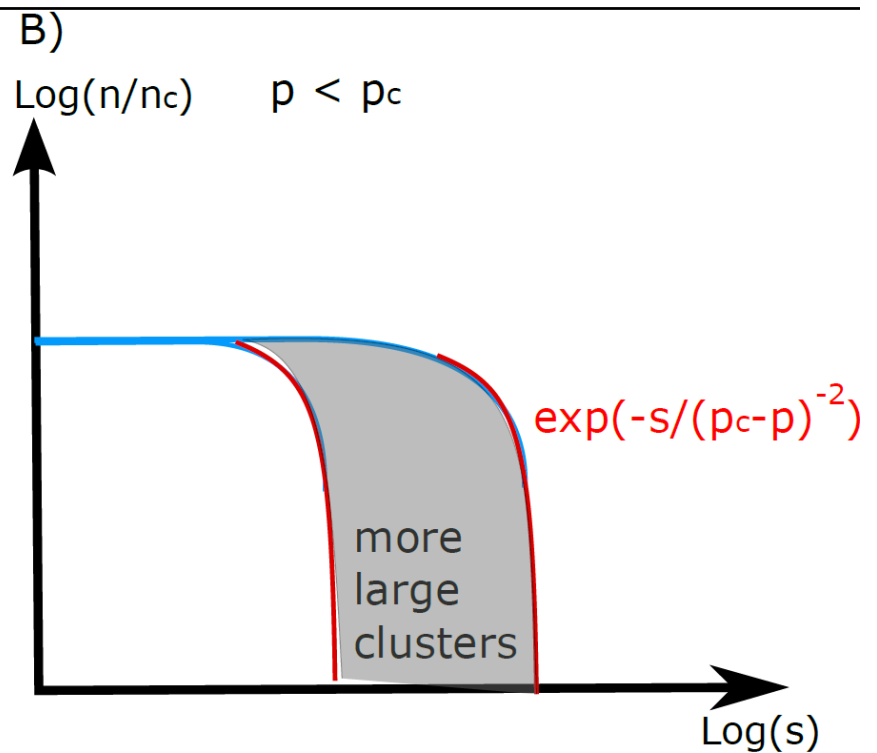
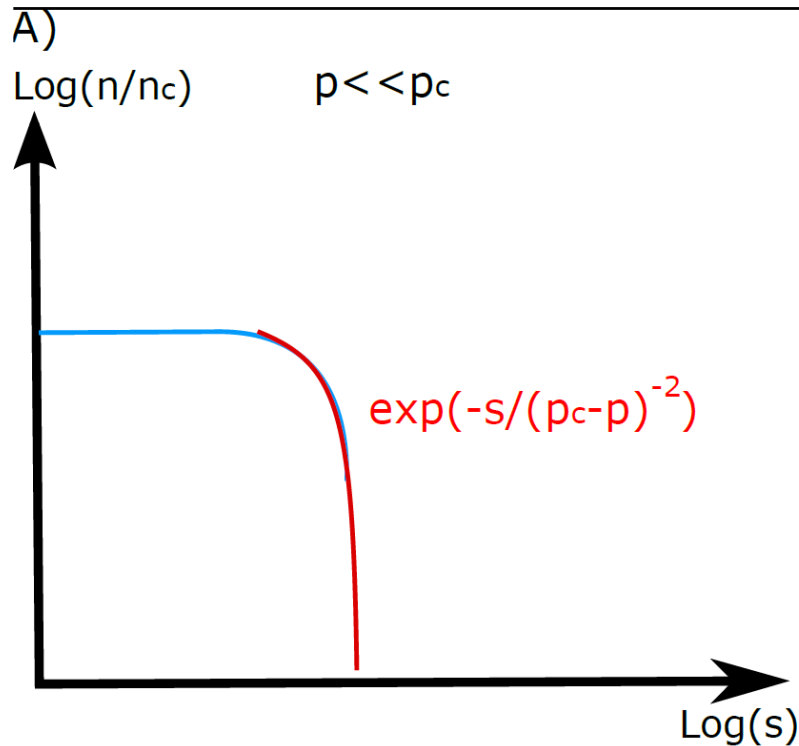
$$\sim \left(\frac{1-p}{1-p_c} \right)^2 \cdot \exp \left(s \cdot \ln(1 - 4 \cdot (p - p_c)^2) \right)$$

$$\sim \left(\frac{1-p}{1-p_c} \right)^2 \cdot \exp \left(-\frac{s}{s_{p-p_c}} \right)$$

$$s_{p-p_c} \equiv \frac{-1}{\ln(1 - 4 \cdot (p - p_c)^2)} \sim \frac{1}{4} \cdot (p - p_c)^{-2} \propto |p - p_c|^{-1/\sigma}$$

Dividing $n(p)$ with $n(p_c)$ gives cutoff

(but not what is below it (no info about $p < p_c$)):



Mini question: Exponent 2 means that if $p-p_c$ halved, then cutoff moves ??

Mini tutorial

Change coordination number from $z = 3$ to $z = 4$ and recalculate the above scaling of maximal cluster size with p approaching p_c (how does relation between t and s change?)

$$n_s(p) = g_{s,2+s} \cdot p^s \cdot (1-p)^{2+s} \quad \rightarrow \quad \text{????}$$

$t=2+s \rightarrow t=2+2s$ (for each s , add 2 to t)

...argue what that does to:

$$\begin{aligned} \frac{n_s(p)}{n_s(p_c)} &= \left(\frac{1-p}{1-p_c} \right)^2 \left(\frac{p}{p_c} \cdot \frac{1-p}{1-p_c} \right)^s \\ &= \left(\frac{1-p}{1-p_c} \right)^2 \exp \left(s \cdot \ln \left(\frac{p}{p_c} \cdot \frac{1-p}{1-p_c} \right) \right) \end{aligned}$$

Assume overall cluster size distribution:

$$n_s(p) = n_s(p_c) \cdot \exp(-s/s_{p_c-p}) \propto s^{-\tau} \cdot \exp(-s/s_{p_c-p})$$

Cluster size distribution *selecting a random occupied site*

$$p(s) = s n_s(p) / \sum_{s=1}^{\infty} s^1 n_s(p)$$

Where the denominator is independent of τ when $\tau > 2$
(lower sum dominated by the lower end).

Average cluster when *selected from an occupied site*:

$$\begin{aligned} S(p) &\propto \sum_{s=1}^{\infty} s^2 n_s(p) / \sum_{s=1}^{\infty} s^1 n_s(p) \\ &\propto \sum_{s=1}^{\infty} s^2 n_s(p) \propto \int_1^{\infty} s^{2-\tau} \cdot \exp(-s/s_{\Delta}) \cdot ds \end{aligned}$$

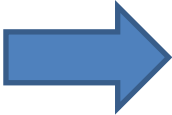
$S(p)$ governed by upper end when $\tau < 3$ (can integrate down to zero):

$$\approx s_{\Delta}^{3-\tau} \cdot \int_0^{\infty} z^{2-\tau} \cdot \exp(-z) \cdot dz = \text{const} \cdot s_{\Delta}^{3-\tau} \propto (p_c - p)^{-1}$$


 $S(p) \propto |p_c - p|^{-\gamma}$

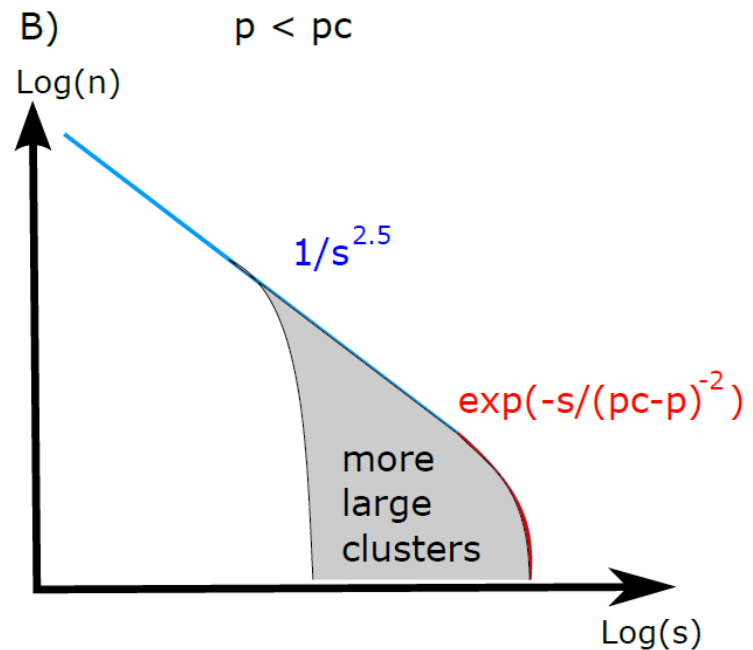
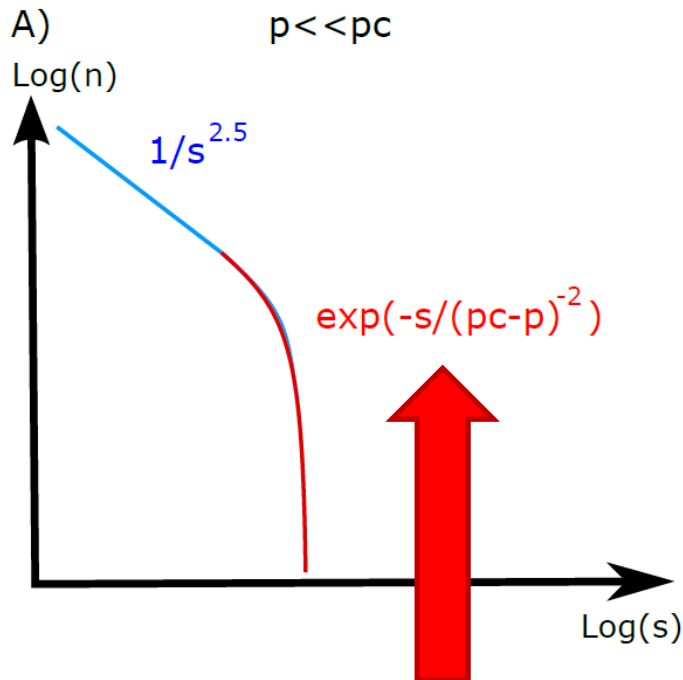
Using $s_{\Delta} \propto (p - p_c)^{-2}$

We get $2\tau - 6 = -1 \rightarrow \tau = \frac{5}{2}$

 $n_s(p) \propto \frac{1}{s^{5/2}} \cdot \exp(-s \cdot |p - p_c|^2)$

Cluster sizes: a power law with a cutoff

$$n_s(p) \propto \frac{1}{s^{5/2}} \cdot \exp(-s \cdot |p - p_c|^2)$$



*Could often be different function, e.g:
 $\exp(-(s/(p_c-p)^2)^4)$*

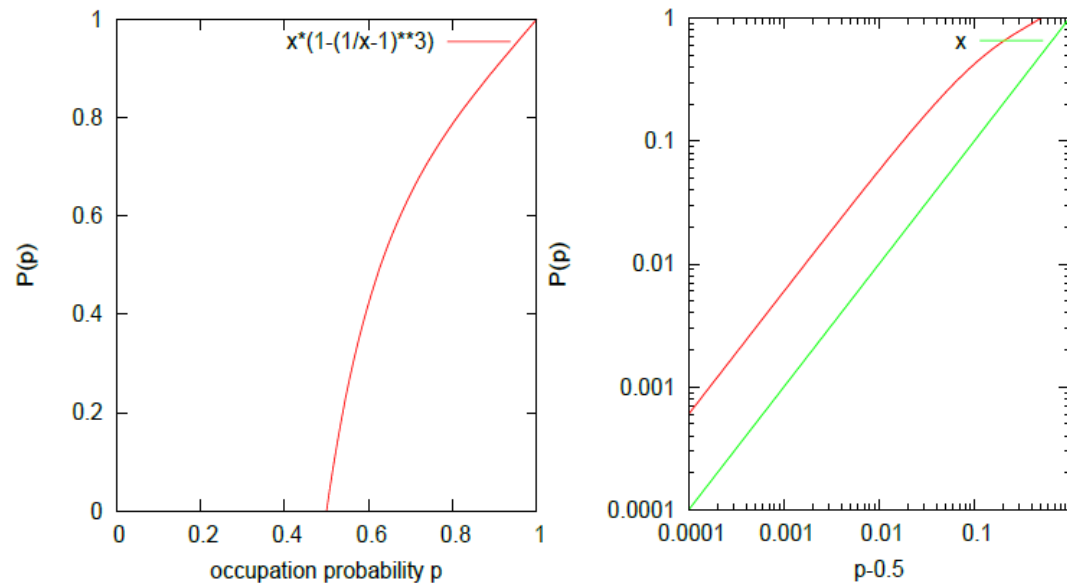
Mini tutorials:

- Select a random point. If point on a cluster then count cluster size. What is the distribution of the thereby sampled cluster sizes?

$$n_s(p) = n_s(p_c) \cdot \exp(-s/s_{p_c-p}) \propto s^{-\tau} \cdot \exp(-s/s_{p_c-p})$$

→?

Another exponent: Density of infinite cluster above p_c



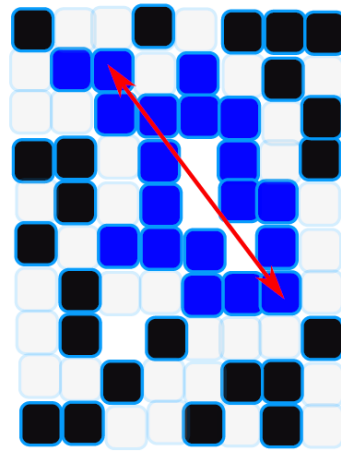
$$P(p) = (p - p_c)^\beta$$

$\beta = 1$ for Bethe lattice, $= 0.14$ for 2-dim perc.

A large exponent means that density is smaller !
..i.e. smaller density at higher dimensions

Correlation length

2 dimensional percolation:



R_{corr} =correlation length

(length of largest clusters)

$$=(p_c - p)^{-1.333}$$

Probability that 2 points separated by distance r are in same cluster

$$g(r) \propto \exp(-r/R_{corr})$$

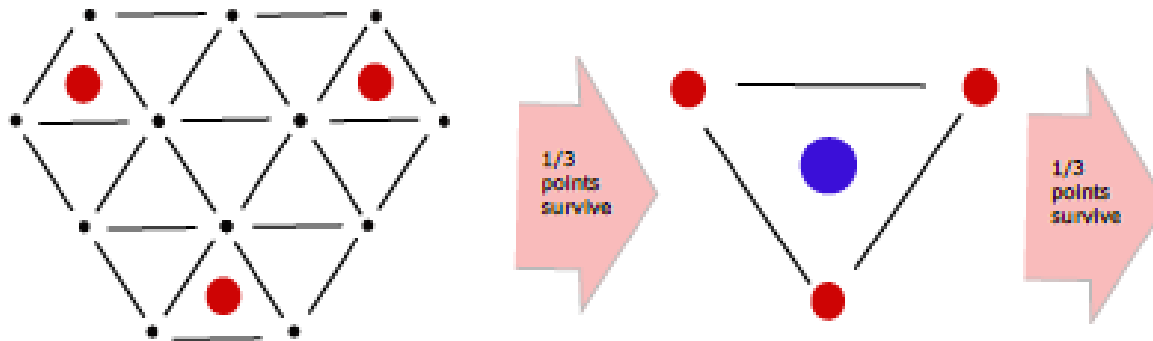
$$R_{corr} \propto |p - p_c|^{-\nu}$$

Smaller exponent means shorter correlation length, ..
In higher dimension the correlation is shorter.

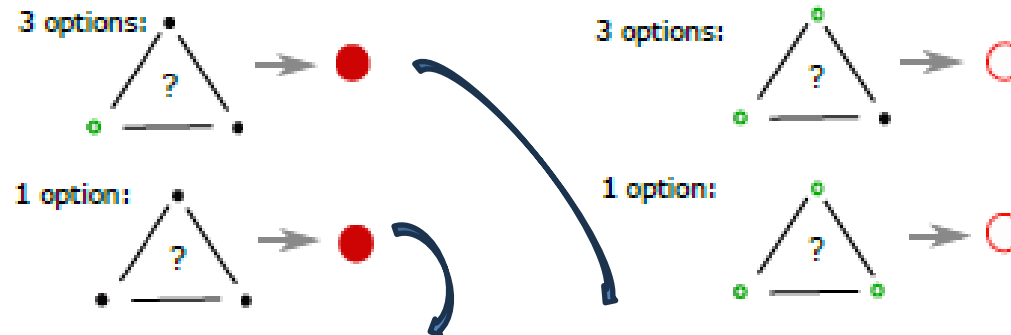
Renormalization $\rightarrow$ V

Consider percolation on a triangular lattice A):
Coarse grain as below with transformation as in B):

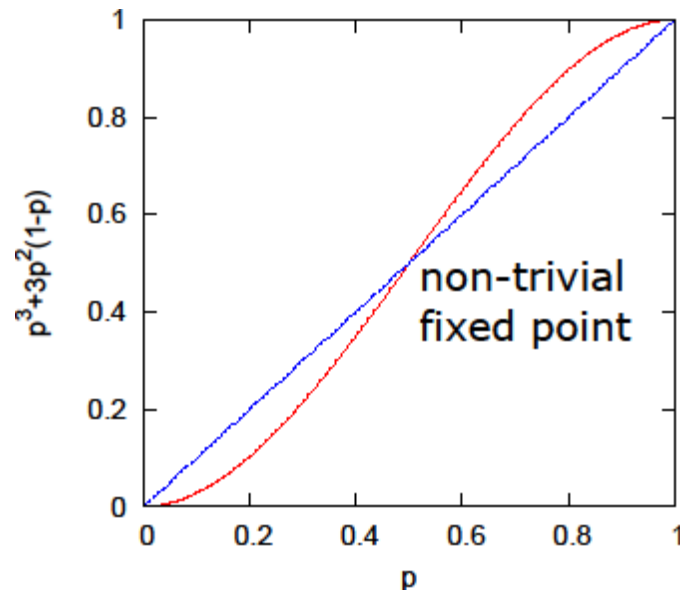
A) Renormalization of triangular lattice, close to p_c :



B) Renormalization transformation



$$p' = p^3 + 3p^2(1 - p)$$



$$p' = p^3 + 3p^2(1 - p)$$

$$\Rightarrow p' = p \Rightarrow p = 1/2$$

Unstable fixed point,
Unstable because it is critical.

$$R_{corr} = |p_c - p|^{-\nu} \Rightarrow R'_{corr} = |p_c - p'|^{-\nu}$$

Where $R'_{corr} = R_{corr}/b$ $b*b=3$, or $b = \sqrt{3}$

$$|p_c - p|^{-\nu}/b = |p_c - p'|^{-\nu} \Rightarrow$$

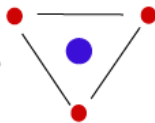
$$b = (dp'/dp)^{\nu} \Rightarrow$$

$$\nu = \frac{\ln(b)}{\ln([dp'/dp]_{p=1/2})} = \frac{\ln(\sqrt{3})}{\ln(3/2)} = 1.355$$

$$p' = 3p^2 - 2p^3 \rightarrow dp' = 6p - 6p^2 = 3 - 3/2 \text{ at } p = 1/2$$

Where $R'_{corr} = R_{corr}/b$.

$b^*b=3$, or $b = \sqrt{3}$

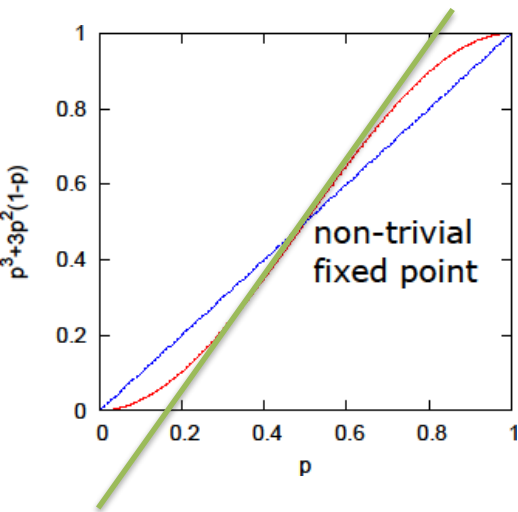


$$|p_c - p|^{-\nu}/b = |p_c - p'|^{-\nu} \Rightarrow$$

$$b = (dp'/dp)^\nu \Rightarrow$$

$$\nu = \frac{\ln(b)}{\ln([dp'/dp]_{p=1/2})} = \frac{\ln(\sqrt{3})}{\ln(\sqrt{3}/2)} = 1.355$$

$\nu = 4/3$
Actual value





$$dp'/dp = -6p^2 + 6p = -6/4 + 6/2 = 3/2$$


Central exponents

dM/dh or $d(\text{order parameter})/dp$

Exponent	Ising 2 dim	Perc. 2-dim.	Perc. 3-dim.	Perc. Bethe lattice
γ	7/4	43/18	1.8	1
β	1/8	5/36	0.41	1
σ		36/91	0.44	1/2
τ		187/91	2.19	5/2
ν	1	4/3	0.88	1/2

$$R_{\text{corr}} = 1/(T - T_c)$$

$$\text{Magnetization} = (T_c - T)^{0.125}$$

$$R_{\text{corr}} = 1/(p_c - p)^{1.34}$$

$$\text{Density largest cluster} = (p - p_c)^{0.14}$$

...in the game of being around the critical point between order and disorder

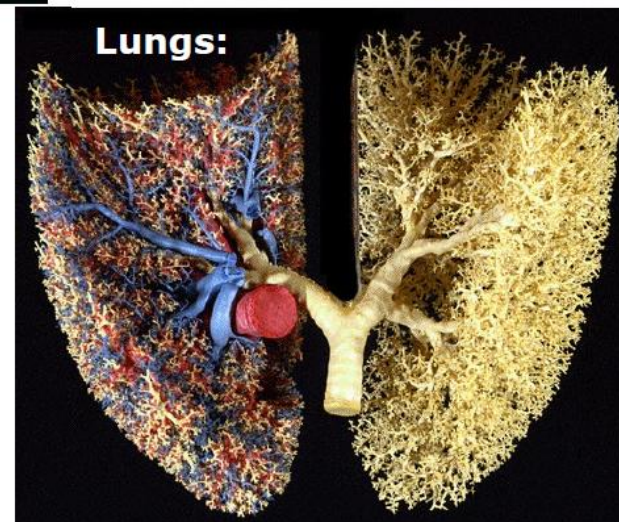
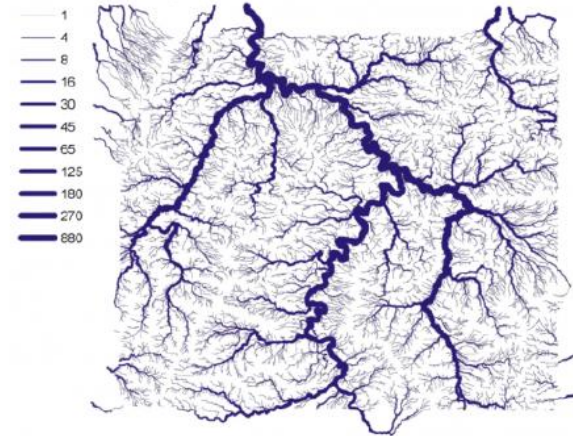
Fractals:

Fractals are everywhere:

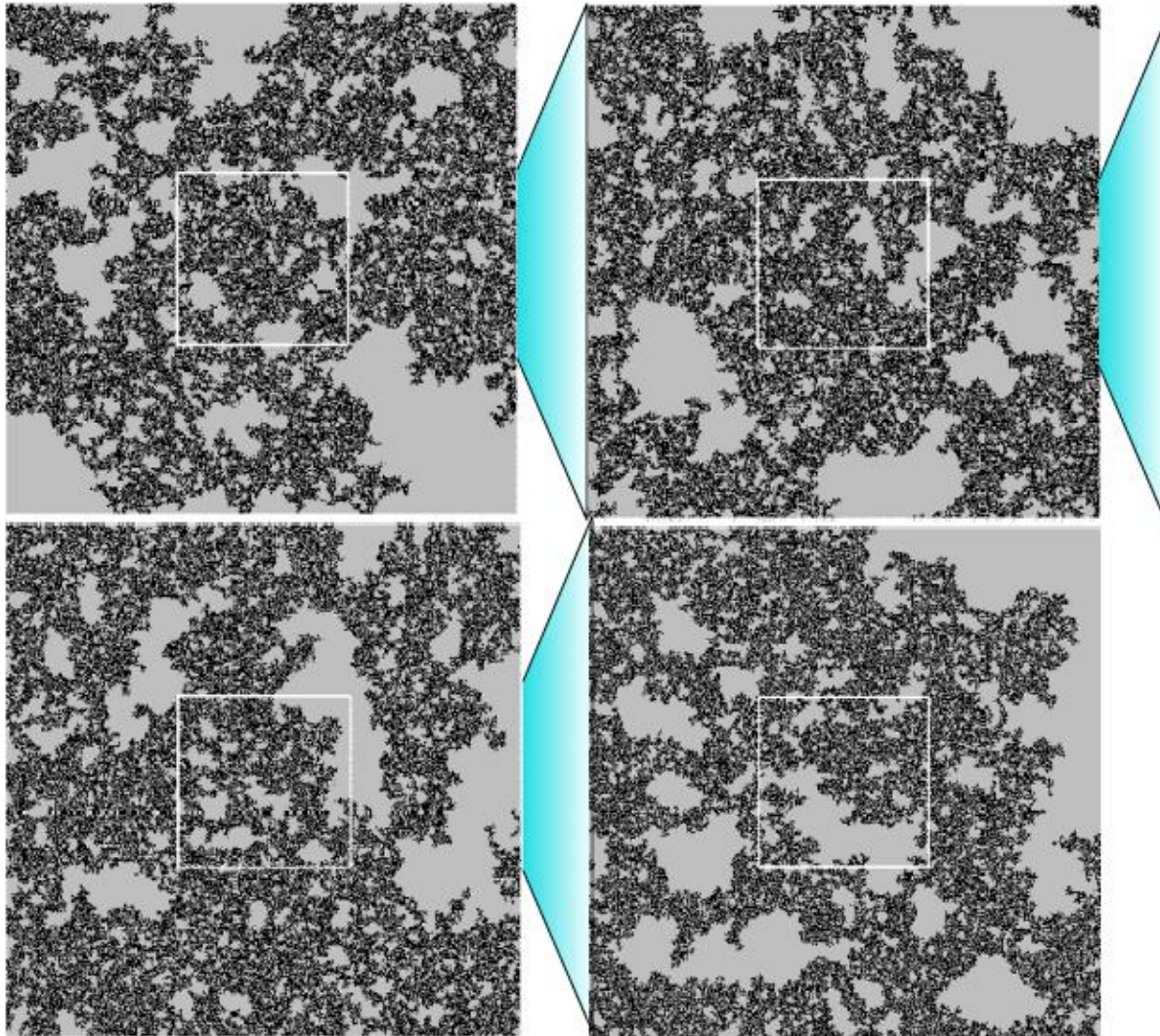


River networks

Contributing Area (km²)



Fractal Dimension (of largest cluster)

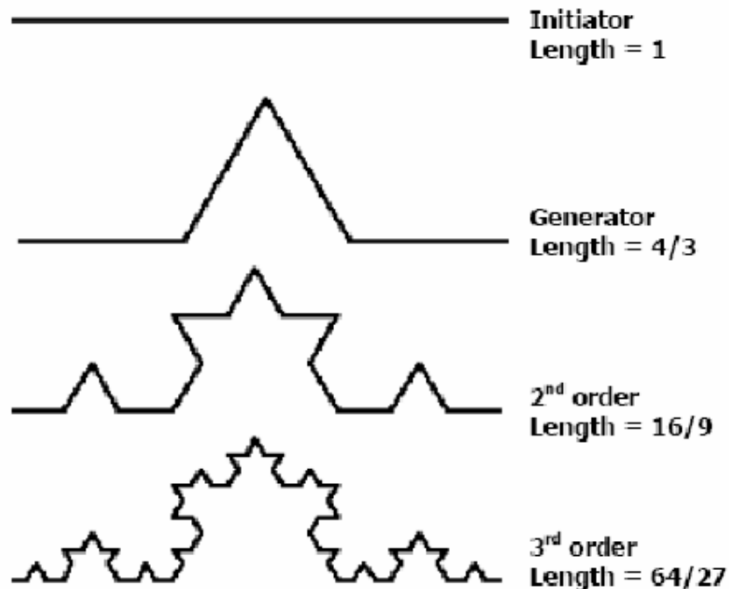


How many boxes do I need to cover the object, in proportion to Sizes of boxes... The more boxes the higher dimension

Fractal dimension

Number boxes
To cover
object

$$D = \lim_{\epsilon \rightarrow 0} \left(\frac{\log(N(\epsilon))}{\log(1/\epsilon)} \right)$$



Make epsilon=1/3

Then need 4 times more boxes

.....

For Each step....

.....

After t steps:

$N_{\text{boxes}}(\epsilon) = 4^t$

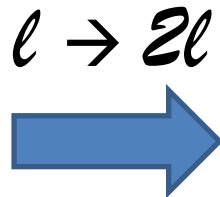
Length of box = $(1/3)^t$

...

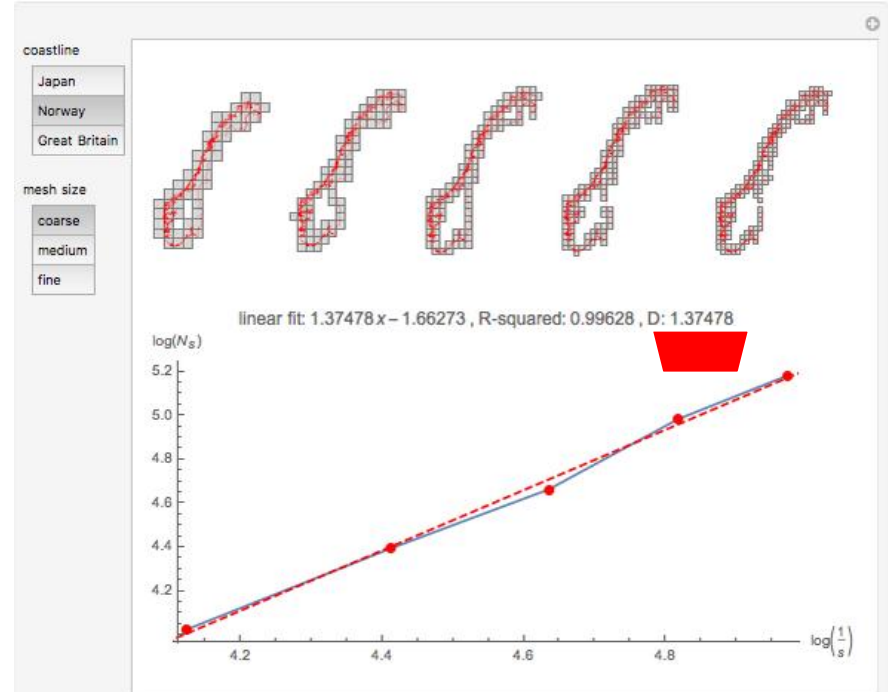
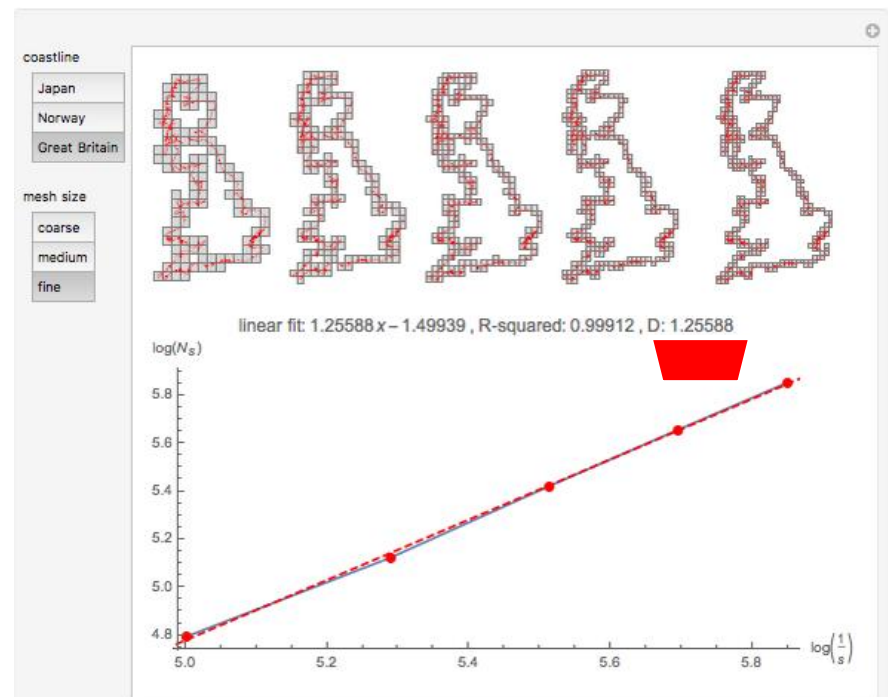
$D = \log(4^t) / \log(3^t) = \log(4) / \log(3) = 1.33$

Fractal Dimension

$$M(l) \propto l^D$$



Dimension of coastlines:



NB: Notice limited scaling regime,
Always check regime of scaling
when discussing exponents.

Percolation: Dimensions & exponent constraints

Below p_c :

$$M \propto (p_c - p)^{-1/\sigma} \quad (\text{cutoff mass})$$

$$R_{corr} \propto (p_c - p)^{-\nu} \Rightarrow (p_c - p) \propto R_{corr}^{-1/\nu} \quad (\text{cutoff length})$$

$$M \propto R_{corr}^{1/\nu\sigma}$$

$$D = \frac{1}{\nu\sigma} = \frac{91}{48} \approx 1.90$$



Exponent	Ising 2 dim	Perc. 2-dim.	Perc. 3-dim.	Perc. Bethe lattice
γ	7/4	43/18	1.8	1
β	1/8	5/36	0.41	1
σ		36/91	0.44	1/2
τ		187/91	2.19	5/2
ν	1	4/3	0.88	1/2

Above p_c :

$$M \sim R_{corr}^d P(p) \propto R_{corr}^d (p - p_c)^\beta \propto R_{corr}^{d-\beta/\nu}$$

$$D = d - \beta/\nu \quad (\text{d=2 in 2-dimension})$$

Should(?) match each other at p_c :

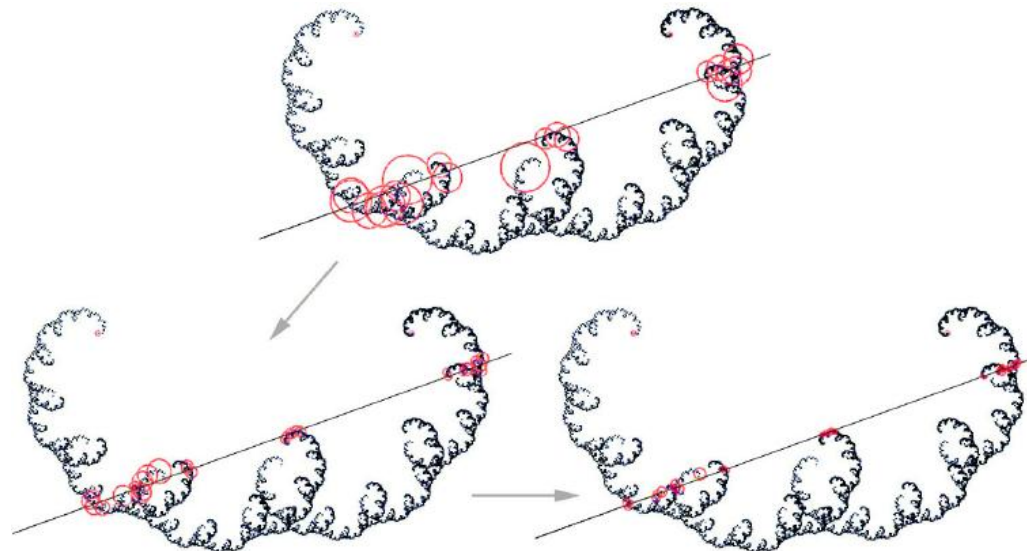
$$\nu = (\beta + 1/\sigma)/2$$

Dimension Equation no. 1:

$$D(A \text{ and } B) = D(A) + D(B) - D_{space}$$

For example 2 random lines in a 2 dimensional space $\rightarrow D(\text{intersect})=0$

For example 2 random planes in a 3 dimensional space $\rightarrow D(\text{intersect})=1$



Red circles
Represent
Boxes of still
Smaller size.
D measured
By counting
boxes

$$P(A) = \frac{(1/\epsilon^{D_A})}{(1/\epsilon^{D_{space}})}$$

$$P(B) = \frac{(1/\epsilon^{D_B})}{(1/\epsilon^{D_{space}})}$$

Independence of A & B →

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

$$\frac{\epsilon^{D_{space}}}{\epsilon^{D(A \text{ and } B)}} = \frac{\epsilon^{D_{space}}}{\epsilon^{D(A)}} \cdot \frac{\epsilon^{D_{space}}}{\epsilon^{D(B)}}$$

For all small epsilons:

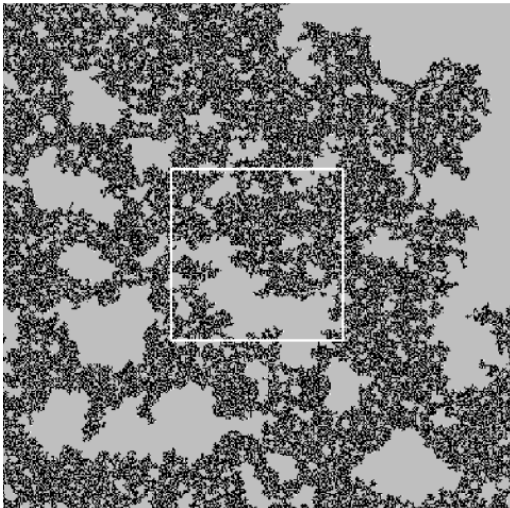
$$D(A \text{ and } B) = D(A) + D(B) - D_{space}$$

Mini tutorials:

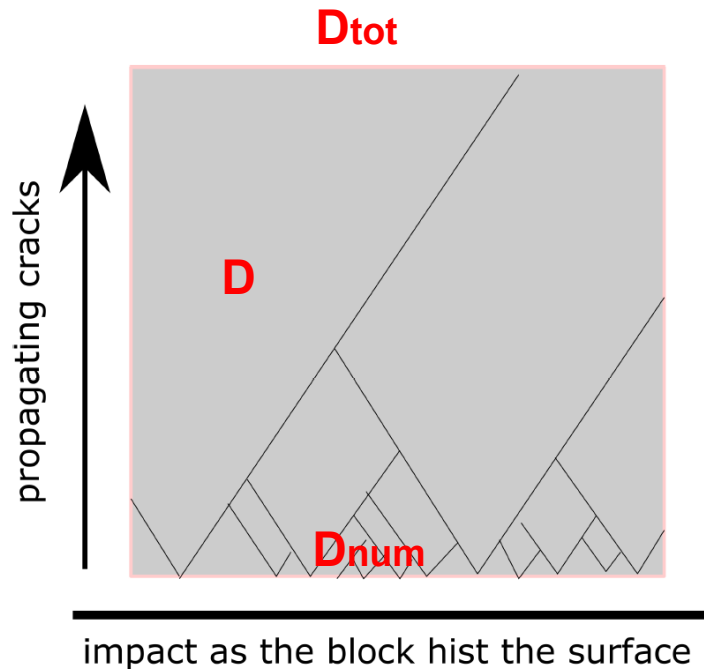
$D=1.90$

Mini tutorial: Calculate the dimension of the intersection between a line and the infinite cluster in two-dimensional percolation at the critical point.

Mini tutorial: Assume the 1.5 dimensional coastline of Norway comes about as the intersection of a 2 dimensional water surface with a mountain range with dimension D . What is D of this “rough” mountain range?



Dimension Equation no. 2:



$$L^{D_{num}} \cdot \int_1^{L^D} s \cdot n(s) ds = L^{D_{tot}}$$

$$L^{D_{num}} \cdot [s^{2-\tau}]_1^{L^D} = L^{D_{tot}}$$

$$L^{D_{num} + (2-\tau)D} = L^{D_{tot}}$$

$$D_{num} + (2 - \tau) \cdot D = D_{tot}$$

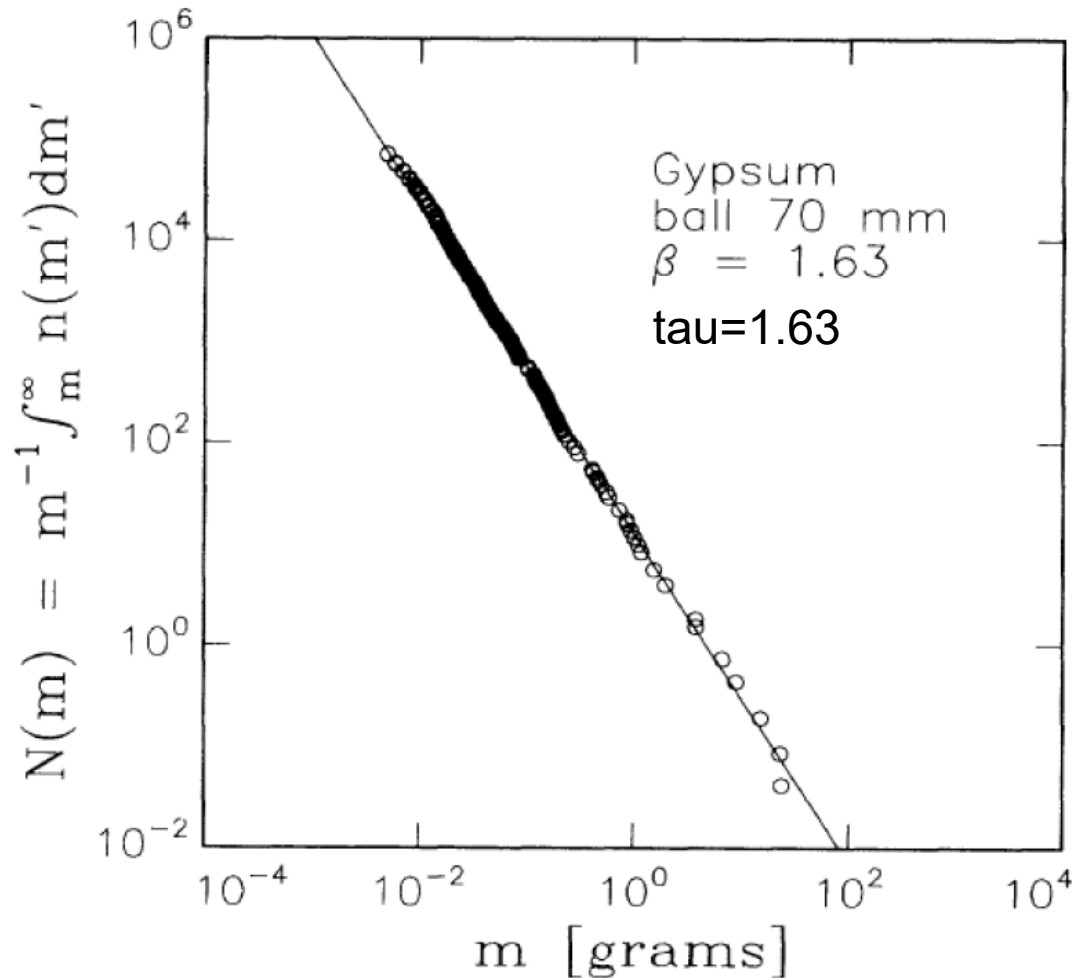
$$\tau = 2 - \frac{D_{tot} - D_{num}}{D},$$

Under the condition that integral is divergent at upper end!!!

$$D_{tot}=D=2, D_{num}=1 \rightarrow \tau=3/2=1.5$$

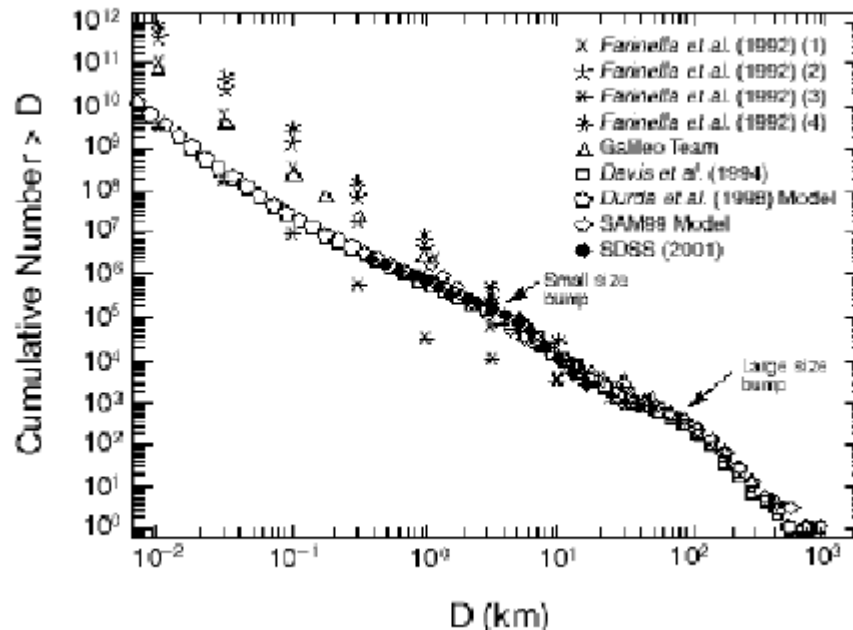
$$D_{tot}=D=3, D_{num}=2 \rightarrow \tau=5/3=1.67$$

Experiment (gypsum fragmentation)



....Amazingly correct agreement given simplification

Meteor size distribution



Mini Tutorial:

$Prop(>Diam) = 1/Diam^{2.2}$

What is cumulative

Mass Distribution:

$P(>M) = Prop(>M^{1/3}) \dots ?$

Mini tutorial:

If a Diam > 10km meteor typically hits earth once per 60.000.000 years.

What is then the risk for a meteor >1km,
and for a meteor >100m?

$$\tau = 2 - \frac{D_{tot} - D_{num}}{D}$$

Not true when $\tau=2.05>2$!!

Then instead: $\tau = 1 + \frac{D_{num}}{D}$ Where $D=1.9$,
 $D_{num}=D_{tot}=2$

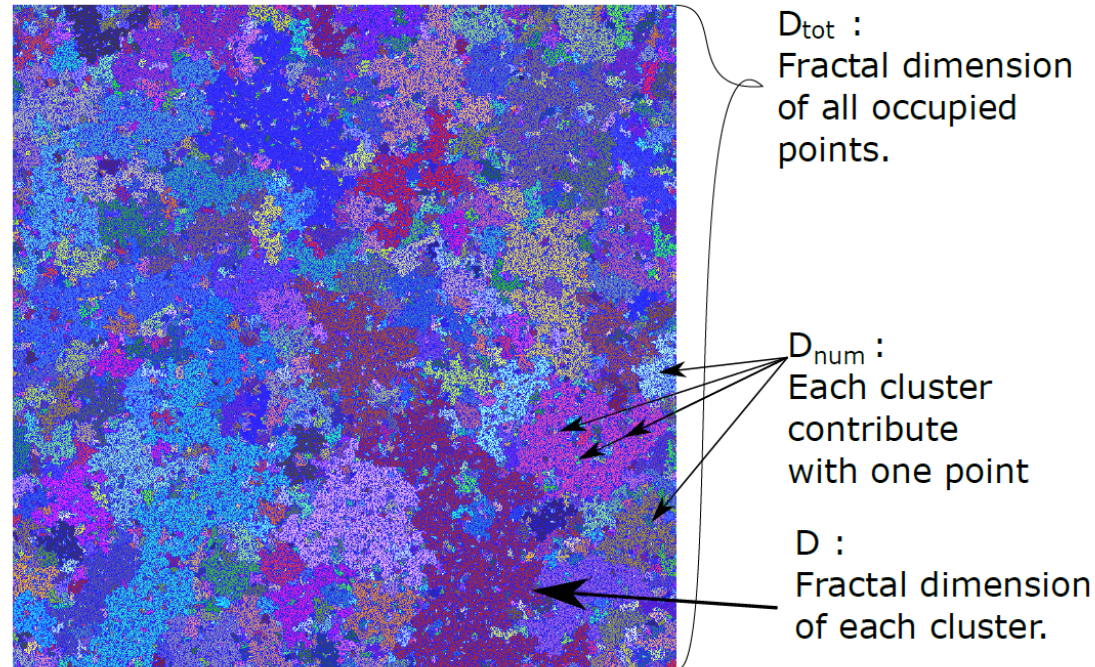
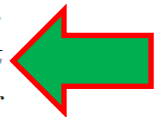


Figure 1.22: **Dimensions for cluster sizes.** A set of clusters can be assigned different fractal dimensions, which together can be used to calculate cluster sizes. Notice that $D_{num} = 2$ for percolation because there is a finite chance that each point belong to a cluster of size 1. This chance is equal $p \cdot (1 - p)^4$ as the size one cluster demand that the 4 nearest neighbors need to be unoccupied.



Directed Percolation

Life spreads, and death is an absorbing state:

Equation perspective:

$$\frac{dn}{dt} = \frac{d^2 n(x, t)}{dx^2} + b \cdot n(x, t) - d \cdot n(x, t)^2 + \eta(x, t)$$

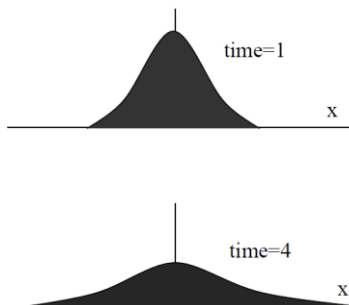
$$\langle \eta(x, t) \eta(x' t') \rangle = n(x, t) \delta(x - x') \delta(t - t')$$

The various terms:

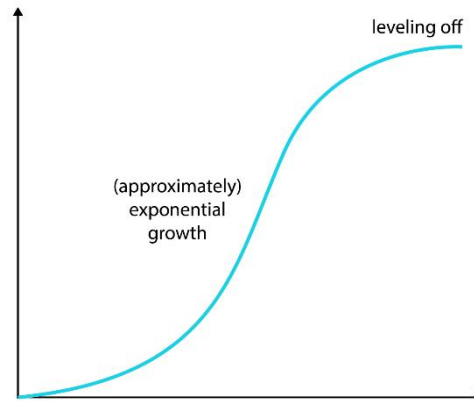
$$\frac{dn}{dt} = \frac{d^2 n(x, t)}{dx^2} + b \cdot n(x, t) - d \cdot n(x, t)^2 + \eta(x, t)$$



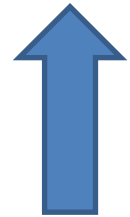
Diffusion
Term
(life tend
to spread)



Life
replicates



There is
limits
To growth



Fate
is
stochastic



Mini tutorial:

$$\frac{dn}{dt} = \frac{d^2n(x,t)}{dx^2} + b \cdot n(x,t) - d \cdot n(x,t)^2 + \eta(x,t)$$

?

- What could be a biological reason for the $-n^2$ term in above equation

$$\langle \eta(x,t) \eta(x',t') \rangle = n(x,t) \delta(x - x') \delta(t - t')$$

?

- What would happen without the $n(x,t)$ factor in above noise size/correlation

Directed percolation,

detailed discrete rule:

Discrete perspective: Synchronous update.

Each site can take +1 or 0.

At each timestep:

any site that have a neighbor in state 1

update to value 1 with probability p .

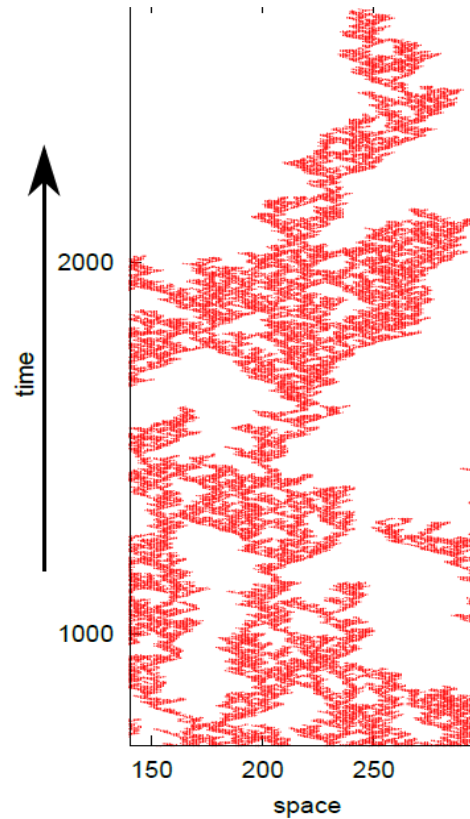
Else it take value=0.

Any state with only 0 among itself and it's neighbors is set to 0

Directed percolation close but below p_c

Spreading distance: $1/|p_c - p|^{1.1}$.

Here we start with all points live at time=0, and then propagate forwards $p < p_c$, leading to eventual death of all live sites,



Survival time:

$$T(p) \propto 1/|p_c - p|^{1.7}$$

Density of life:

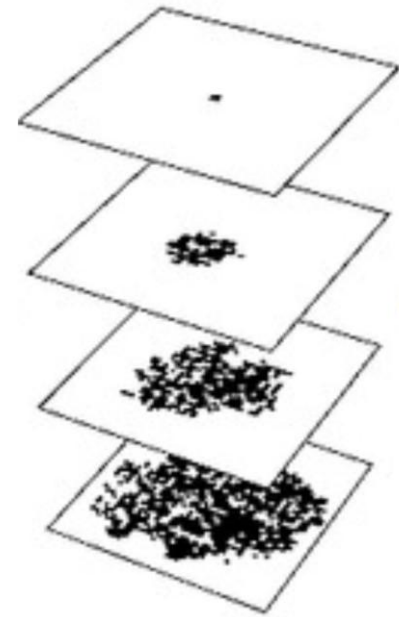
$$\rho \propto |p - p_c|^\beta, \beta = 0.27.$$

Mini tutorials

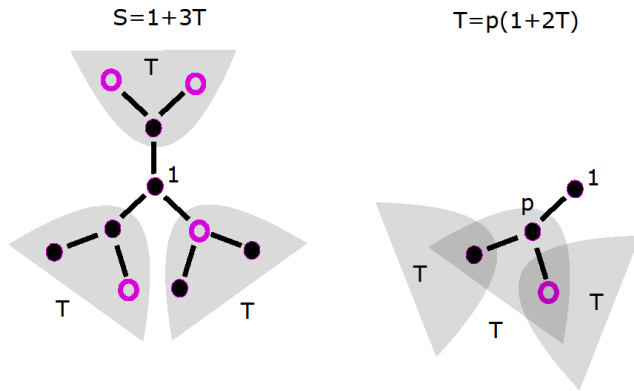
- What does a smaller β means in:
 $\rho = P(p) = (p - p_c)^\beta$



Exponent	Mean-field	1-d DP	2-d DP
β	1	0.276	0.58
$\nu_\perp$	1/2	1.097	0.73
$\nu_\parallel$	1	1.734	1.2
δ	1/2	0.159	0.45



Recapitulation 18/9



Average cluster size starting from occupied site =

$$S(p) = 1 + 3T = \frac{1 + p}{2(p_c - p)} \propto (p_c - p)^{-1}$$

$$n_s(p) = \sum_t g_{s,t} p^s (1-p)^t$$

$$n_s(p) = g_{s,2+s} \cdot p^s \cdot (1-p)^{2+s}$$

$$s_{p-p_c} \equiv \frac{-1}{\ln(1 - 4 \cdot (p - p_c)^2)} \sim \frac{1}{4} \cdot (p - p_c)^{-2} \propto |p - p_c|^{-1/\sigma}$$

$$n_s(p) = n_s(p_c) \cdot \exp(-s/s_{p_c-p}) \propto s^{-\tau} \cdot \exp(-s/s_{p_c-p})$$

$$S(p) \propto \frac{\sum_{s=1}^{\infty} s^2 n_s(p)}{\sum_{s=1}^{\infty} s^1 n_s(p)}$$

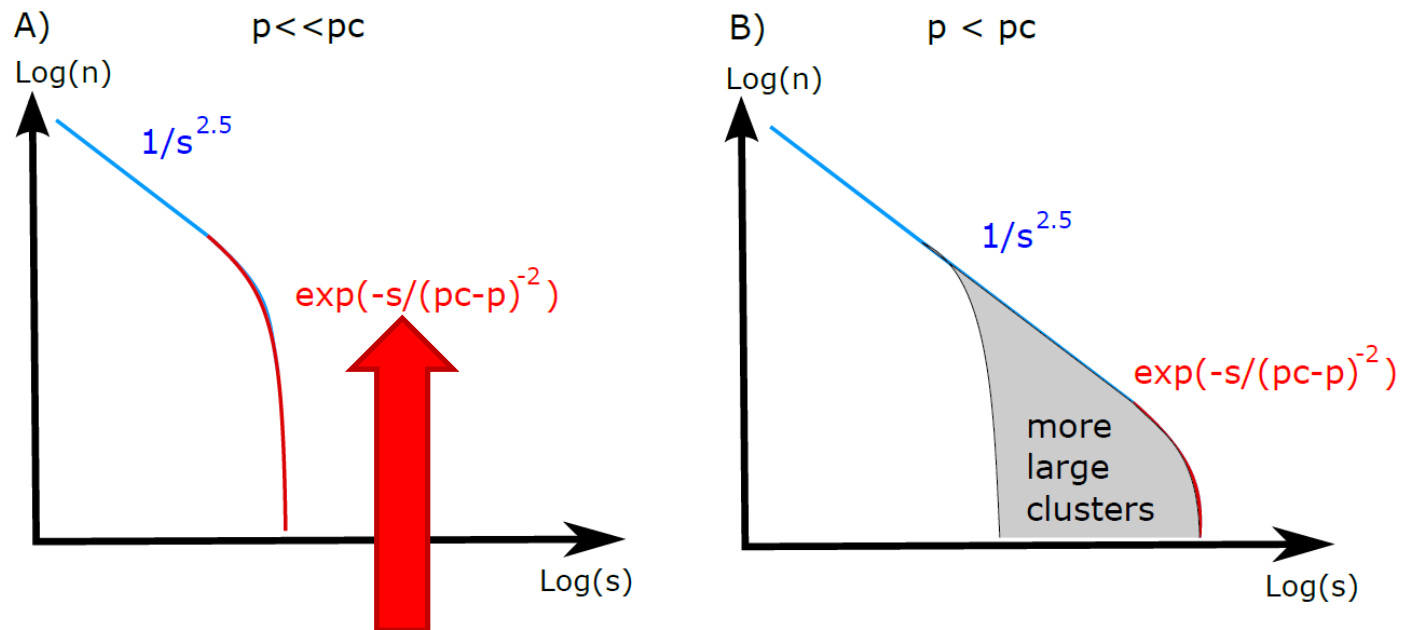
$$\propto \sum_{s=1}^{\infty} s^2 n_s(p) \propto \int_1^{\infty} s^{2-\tau} \cdot \exp(-s/s_{\Delta}) \cdot ds \approx s_{\Delta}^{3-\tau} \cdot \int_0^{\infty} z^{2-\tau} \cdot \exp(-z) \cdot dz = \text{const} \cdot s_{\Delta}^{3-\tau} \propto (p_c - p)^{-1}$$

$$n_s(p) \propto \frac{1}{s^{5/2}} \cdot \exp(-s \cdot |p - p_c|^2)$$

Recapitulation, September 18 (2024):

Bethe Lattice exponents for cluster sizes in percolation:

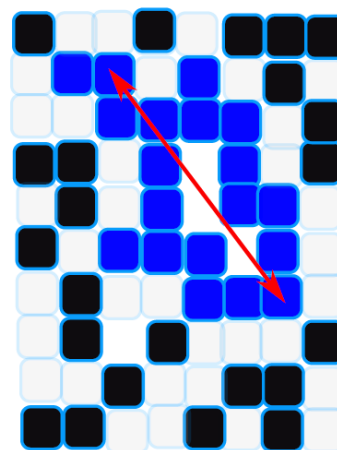
$$n_s(p) \propto \frac{1}{s^{5/2}} \cdot \exp(-s \cdot |p - p_c|^2)$$



*Could often be different function, e.g:
 $\exp(-(s/(p_c-p)^2)^4)$*

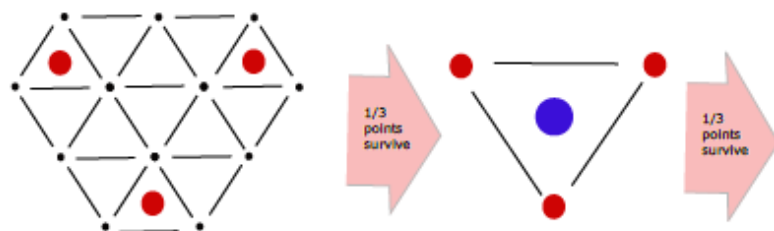
Recapitulation, Renormalization

2 dimensional percolation:

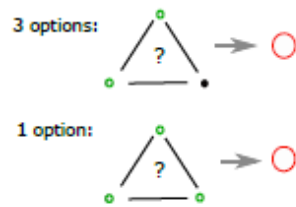
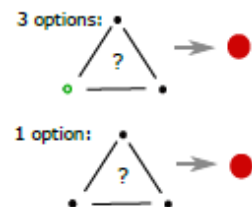


Rcorr=correlation length
(length of largest clusters)
 $= (p_c - p)^{-1.333}$

A) Renormalization of triangular lattice, close to p_c :



B) Renormalization transformation



$$p' = p^3 + 3p^2(1 - p)$$



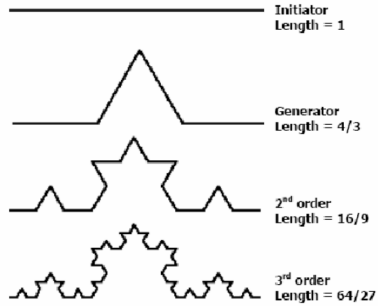
$$p = 1/2$$

$$|p_c - p|^{-\nu} / b = |p_c - p'|^{-\nu} \Rightarrow$$

$$b = (dp'/dp)^\nu \Rightarrow$$

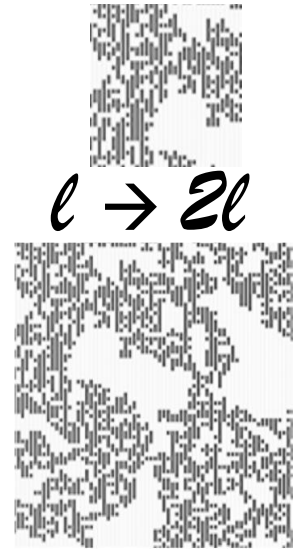
$$\nu = \frac{\ln(b)}{\ln([dp'/dp]_{p=1/2})} = \frac{\ln(\sqrt{3})}{\ln(3/2)} = 1.355$$

Recapitulation, September 18 (2024):



$$D = \lim_{\epsilon \rightarrow 0} \left(\frac{\log(N(\epsilon))}{\log(1/\epsilon)} \right)$$

$$\longleftrightarrow M(l) \propto l^D$$



$$D(A \text{ and } B) = D(A) + D(B) - D_{space}$$

Recapitulation, September 18 (2024):

Percolation: Dimensions & exponent constraints

Below p_c : $M \propto (p_c - p)^{-1/\sigma}$ (cutoff mass)

$R_{corr} \propto (p_c - p)^{-\nu} \Rightarrow (p_c - p) \propto R_{corr}^{-1/\nu}$ (cutoff length)

$M \propto R_{corr}^{1/\nu\sigma}$ $D = \frac{1}{\nu\sigma} = \frac{91}{48} \approx 1.90$

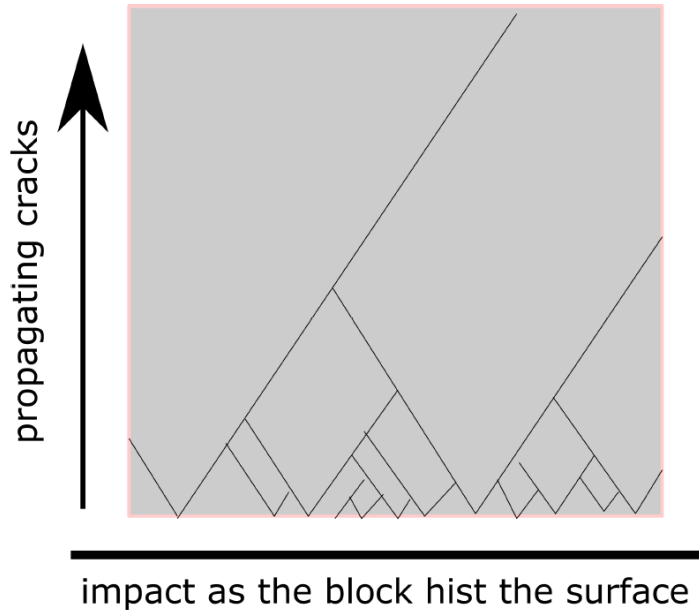


Exponent	Ising 2 dim	Perc. 2-dim.	Perc. 3-dim.	Perc. Bethe lattice
γ	7/4	43/18	1.8	1
β	1/8	5/36	0.41	1
σ		36/91	0.44	1/2
τ		187/91	2.19	5/2
ν	1	4/3	0.88	1/2

Recapitulation, September 18 (2025):

$$\frac{\epsilon^{D_{space}}}{\epsilon^{D(A \text{ and } B)}} = \frac{\epsilon^{D_{space}}}{\epsilon^{D(A)}} \cdot \frac{\epsilon^{D_{space}}}{\epsilon^{D(B)}}$$

$$P(A \text{ and } B) = P(A) \cdot P(B) \quad \longrightarrow \quad D(A \text{ and } B) = D(A) + D(B) - D_{space}$$



$$L^{D_{num}} \cdot \int_1^{L^D} s \cdot n(s) ds = L^{D_{tot}}$$

$$L^{D_{num}} \cdot [s^{2-\tau}]_1^{L^D} = L^{D_{tot}}$$

$$L^{D_{num} + (2-\tau)D} = L^{D_{tot}}$$

$$D_{num} + (2 - \tau) \cdot D = D_{tot}$$

$$\tau = 2 - \frac{D_{tot} - D_{num}}{D},$$

Not true when $\tau=2.05>2$!!

Then instead: $\tau = 1 + \frac{D_{num}}{D}$ Where $D=1.9$,
 $D_{num}=D_{tot}=2$

Directed percolation... equation form:

$$\frac{dn}{dt} = \frac{d^2 n(x, t)}{dx^2} + b \cdot n(x, t) - d \cdot n(x, t)^2 + \eta(x, t)$$



- What could be a biological reason for the $-n^2$ term in above equation

$$\langle \eta(x, t) \eta(x', t') \rangle = n(x, t) \delta(x - x') \delta(t - t')$$



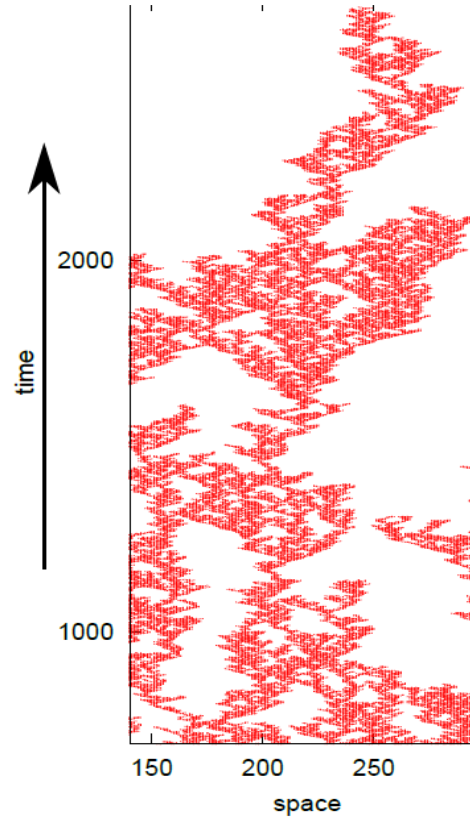
- What would happen without the $n(x, t)$ factor in above noise size/correlation

Recapitulation, September 18 (2025):

Directed percolation close but below p_c

Spreading distance: $1/|p_c - p|^{1.1}$.

Here we start with all points live at time=0, and then propagate forwards $p < p_c$, leading to eventual death of all live sites,



Survival time:

$$T(p) \propto 1/|p_c - p|^{1.7}$$

Density of life:

$$\rho \propto |p - p_c|^\beta, \beta = 0.27.$$

Wolfram nomenclature in 1d:

Binary output $f=0$ or $f=1$

$f(111)$ $f(110)$ $f(101)$ $f(100)$ $f(011)$ $f(010)$ $f(001)$ $f(000)$

128

64

32

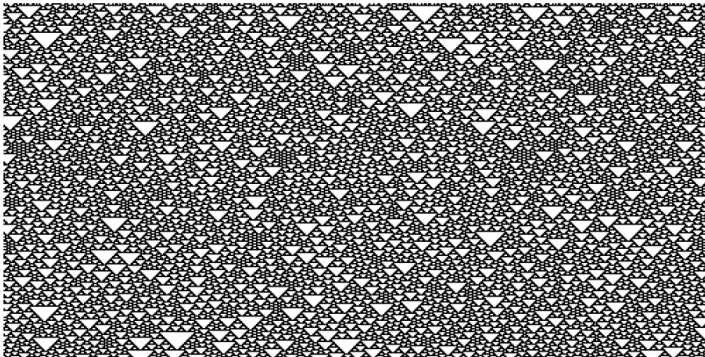
16

8

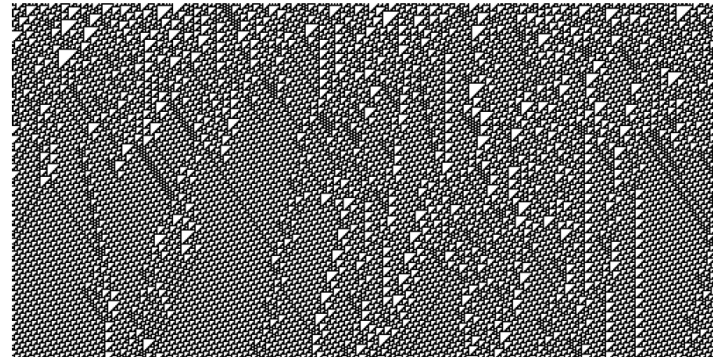
4

2

+1



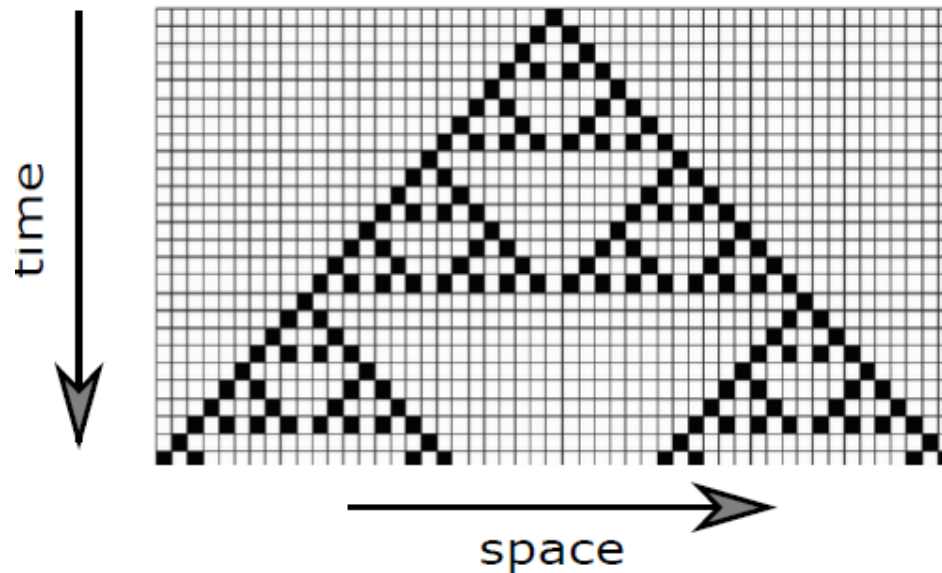
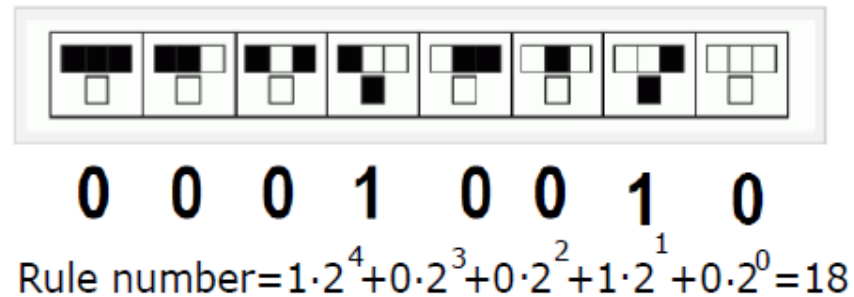
(class 3: rule 126)



(class 4: rule 110)

$$\text{Rule number} = \sum f(\dots) 2^i \quad \text{with } i=1, 2, \dots, 7$$

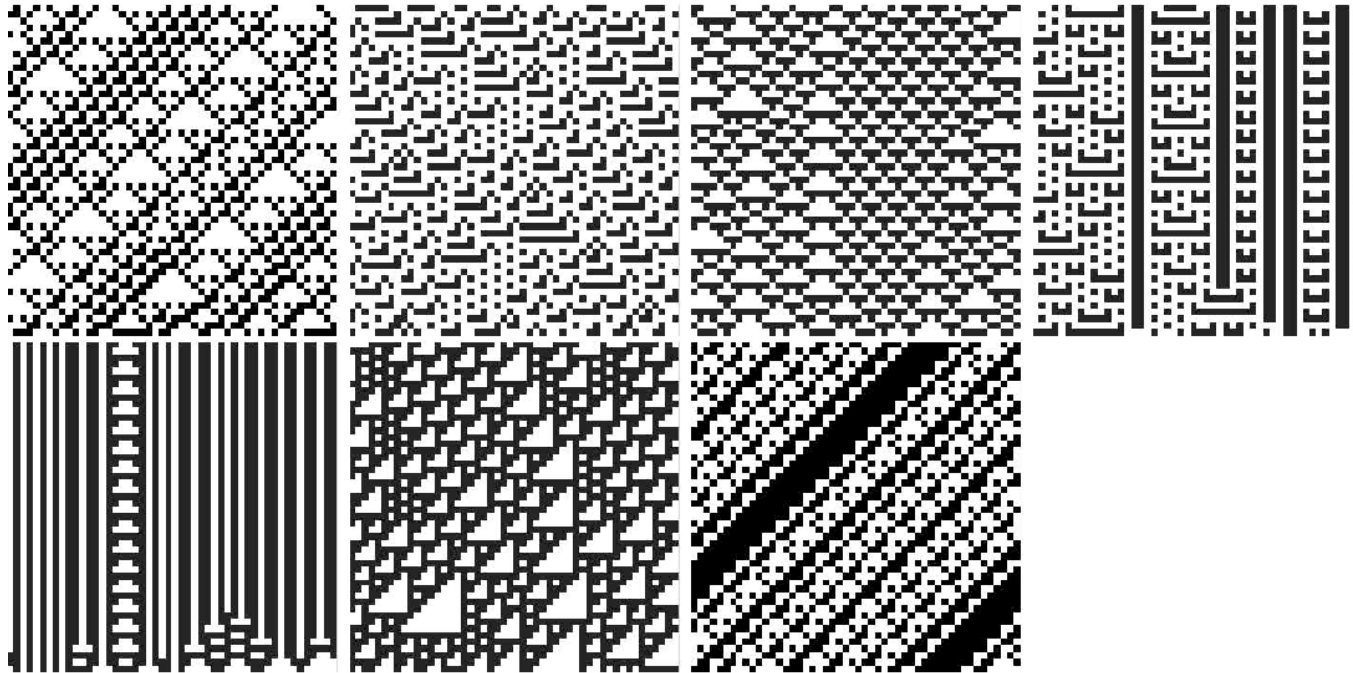
Probabilistic Cellular automata with similarity to DP



One should of course add a probabilistic element, i.e. not all $\rightarrow 1$ works

Other rules:

Fig. 2. Some space-time diagrams obtained with the rules 26, 41, 54, 73, 94, 110, 154 which represent the 7 symmetric-equivalent (see §3.5) hybrid rules.



4.2 Comparison with Wolfram's phenotypic classification

The classification proposed by Wolfram in [Wol84] divides the CA space into 4 classes :

Class W1	evolution of the CA leads to homogeneous stable configuration
Class W2	evolution of the CA leads to a set of separated simple or periodic structures
Class W3	evolution of the CA leads to a chaotic pattern
Class W4	evolution leads to complex localised structures sometimes long-lived

Mini tutorials:

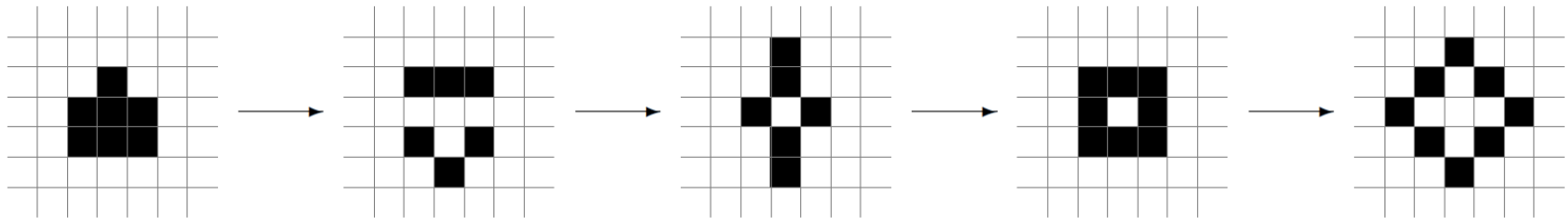
- How would the iterated version of rule 255 looks like?
- How would a probabilistic version of rule 225 plays out?

Cellular Automata

2-dimensional

--game of life:

- A living cell stays alive if and only if there are exactly two or three living cells among the eight surrounding cells. Fewer than two living neighbors causes death by isolation, more than three living neighbors by overcrowding.
- A non-living cell becomes alive if it has precisely three living neighbors — each organism has three parents!



Game of life... snapshot:

